Exercise J-M

1. Let $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$. If $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$, then f(4) is equal to :

[JEE(Main)Jan.2023]

माना $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$ है। यदि $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$ तो f(4) बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{1}{2}(\log_e 17-\log_e 19)$ (B) $\frac{1}{2}(\log_e 19-\log_e 17)$

(C) $\log_e 17-\log_e 18$ (D) $\log_e 19-\log_e 20$

Ans. A

Sol. $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$

Let $x^2=t$

$2xdx=dt$

$\int \frac{dt}{(t+1)(t+3)}$

$=\frac{1}{2}\int \frac{(t+3)-(t+1)}{(t+1)(t+3)}dt$

$=\frac{1}{2}[\ln|t+1|-\ln|t+3|]+\frac{C}{2}$

$=\frac{1}{2}\left[\ln\left|x^2+1\right|-\ln\left|x^2+3\right|\right]+\frac{C}{2}$

$\because \quad f(3)=\frac{1}{2}[\ln 5-\ln 6]$

$\therefore \quad \frac{1}{2}[\ln 5-\ln 6]=\frac{1}{2}[\ln 10-\ln 12]+\frac{C}{2}$

$\Rightarrow \quad C=0$

$\therefore \quad f(x)=\frac{1}{2}\left[\ln\left|x^2+1\right|-\ln\left|x^2+3\right|\right]$

$f(4)=\frac{1}{2}[\ln 17-\ln 19]$

2. If $\int\sqrt{\sec 2x-1}dx=\alpha\log_e\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+\text{constant}$, then $\beta-\alpha$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\int\sqrt{\sec 2x-1}dx=\alpha\log_e\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+\text{constant}$, तो $\beta-\alpha$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $\int\sqrt{\sec 2x-1}dx=\int\sqrt{\frac{1-\cos 2x}{\cos 2x}}dx=\sqrt{2}\int\frac{\sin x}{\sqrt{2\cos^2 x-1}}$

put $\cos x=t\Rightarrow -\sin x\,dx=dt$

$$=-\sqrt{2}\int\frac{dt}{\sqrt{2t^2-1}}=-\ln\left|\sqrt{2}\cos x+\sqrt{\cos 2x}\right|+c=-\frac{1}{2}\ln\left|\cos 2x+\frac{1}{2}+\sqrt{\cos 2x}.\sqrt{1+\cos 2x}\right|+c$$

$\because \beta=\frac{1}{2},\alpha=-\frac{1}{2}\Rightarrow\beta-\alpha=1$

3. If $\int_0^1\left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{\frac{1}{7}}dx=\frac{1}{\ell}(11)^{\frac{m}{n}}$ where l, m, n $\in$ N, m and n are coprime then $l+m+n$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\int_0^1\left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{\frac{1}{7}}dx=\frac{1}{\ell}(11)^{\frac{m}{n}}$ है, जहाँ l, m, n $\in$ N, m एवं n सहअभाज्य हैं, तब $l+m+n$ का मान है। **[JEE(Main)Jan.2023]**

Ans. 63

Sol. $\int\left(x^{20}+x^{13}+x^6\right)\left(2x^{21}+3x^{14}+6x^7\right)^{1/7}dx$

$2x^{21}+3x^{14}+6x^7=t$

$42(x^{20}+x^{13}+x^6)\,dx=dt$

$$\frac{1}{42}\int_0^{11}t^{\frac{1}{7}}dt=\left(\frac{t^{\frac{8}{7}}}{\frac{8}{7}}\times\frac{1}{42}\right)_0^{11}$$

$$\frac{1}{48}\left(t^{\frac{8}{7}}\right)_0^{11}=\frac{1}{48}(11)^{8/7}$$

$l=48,\ m=8,\ n=7$

$l + m + n = 63$

4. $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to : **[JEE(Main)Jan.2023]**

$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{3}$ (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. $\frac{48}{2}\left(\sin^{-1}\left(\frac{2x}{3}\right)\right)_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} = 24\left(\sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}\frac{1}{\sqrt{2}}\right)$

$= 24\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = 24\left(\frac{\pi}{12}\right) = 2\pi$

5. The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}\,dt$ is : **[JEE(Main)Jan.2023]**

फलन $f(x) = \int_0^2 e^{|x-t|}\,dt$ का निम्नतम मान है : **[JEE(Main)Jan.2023]**

(A) 2 (B) e (e – 1) (C) 2 (e – 1) (D) 2e – 1

Ans. C

Sol. For $x > 2$

$f(x) = \int_0^2 e^{x-t}dt$

$= e^x\left(-e^{-t}\right)\Big|_0^2$

$= e^x\left(1 - e^{-2}\right)$

For $x < 0$

$f(x) = \int_0^2 e^{t-x}dt = e^{-x}e^t\Big|_0^2 = e^{-x}\left(e^2 - 1\right)$

For $0 \le x \le 2$

$f(x) = \int_0^x e^{x-t}dt + \int_x^2 e^{t-x}dt$

$= -e^xe^{-t}\Big|_0^x + e^{-x}e^t\Big|_x^2$

$\Rightarrow -e^{x}\left(e^{-x}-1\right)+e^{-x}\left(e^{2}-e^{x}\right)$

$\Rightarrow -1+e^{x}+e^{2-x}-1$

$=e^{2-x}+e^{x}-2$

$$f(x)=\begin{cases} e^{x}\left(1-e^{-2}\right); & x>2 \\ e^{2-x}+e^{x}-2; & 0\le x\le 2 \\ e^{-x}\left(e^{2}-1\right); & x<0 \end{cases}$$

For $x>2$

$f(x)_{min}=e^{2}-1$

For $0\le x\le 2$

$f'(x)=-e^{2-x}+e^{x}=0\Rightarrow e^{x}=e^{2-x}\Rightarrow e^{2x}=e^{2}\Rightarrow x=1$

$f(x)=2e-2=2(e-1)$

For $x<0$

$f(x)_{min}=e^{2}-1$

6. The integral $16\int_{1}^{2}\frac{dx}{x^{3}\left(x^{2}+2\right)^{2}}$ is equal to : **[JEE(Main)Jan.2023]**

समाकलन $16\int_{1}^{2}\frac{dx}{x^{3}\left(x^{2}+2\right)^{2}}$ बराबर है **[JEE(Main)Jan.2023]**

(A) $\frac{11}{6}+\log_{e}4$ (B) $\frac{11}{12}+\log_{e}4$ (C) $\frac{11}{6}-\log_{e}4$ (D) $\frac{11}{12}-\log_{e}4$

Ans. C

Sol. $\int_{1}^{2}\frac{dx}{x^{3}x^{4}}\left(1+\frac{2}{x^{2}}\right)^{2}$

Let $1+\frac{2}{x^{2}}=t\Rightarrow\frac{-4}{x^{3}}dx=dt$

$I=\frac{-1}{4}\int_{3}^{\frac{3}{2}}\left(\frac{t-1}{2}\right)^{2}\frac{dt}{t^{2}}=-\frac{1}{16}\int_{3}^{\frac{3}{2}}\frac{t^{2}-2t+1)}{t^{2}}dt$

$=\frac{-1}{16}\int_{3}^{\frac{3}{2}}\left(1-\frac{2}{t}+\frac{1}{t^{2}}\right)dt$

$=\frac{-1}{16}\left(t-2\ln|t|-\frac{1}{t}\right)_{3}^{\frac{3}{2}}$

$$=\frac{-1}{16}\left(\frac{3}{2}-2\ln\frac{3}{2}-\frac{2}{3}\right)-(3-2)n2-\frac{1}{3}\Bigg)$$

$$=\frac{-1}{16}\left(2\left(\ln 2-\ln\frac{3}{2}\right)-\frac{3}{2}-\frac{1}{3}\right)$$

$$=\frac{-1}{16}\left(2\ln 3-\frac{11}{6}\right)=\frac{-1}{16}\left(\ln 4-\frac{11}{6}\right)=\frac{11-6\ln 4}{16\times 6}$$

$$\frac{11-6\ln 4}{96}$$

7. Let [x] denote the greatest integer $\le x$. Consider the function $f(x) = \max\{x^2, 1+[x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is : **[JEE(Main)Jan.2023]**

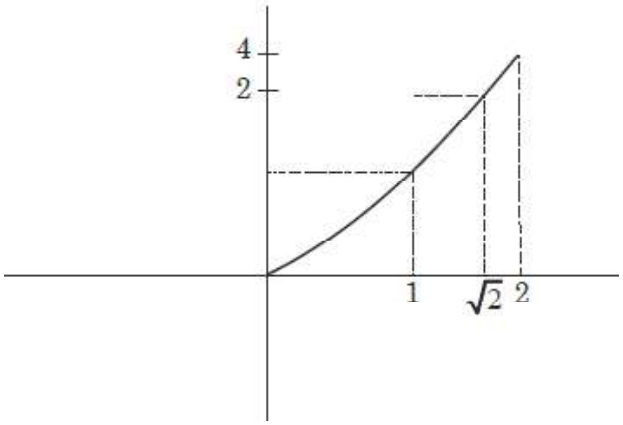
माना [x] महत्तम पूर्णांक $\le x$ है। माना फलन $f(x) = \max\{x^2, 1+[x]\}$ है। तब समाकलन $\int_0^2 f(x)dx$ का मान है :

[JEE(Main)Jan.2023]

(A) $\frac{8+4\sqrt{2}}{3}$ (B) $\frac{5+4\sqrt{2}}{3}$ (C) $\frac{1+5\sqrt{2}}{3}$ (D) $\frac{4+5\sqrt{2}}{3}$

Ans. B

Sol.



$$A=\int_0^1 1.dx+\int_1^{\sqrt{2}} 2dx+\int_{\sqrt{2}}^2 x^2dx$$

$$=1+2\sqrt{2}-2+\frac{8}{3}-\frac{2\sqrt{2}}{3}$$

$\frac{5}{3}+\frac{4\sqrt{2}}{3}$,

8. Let $f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in IR$ be a function which satisfies

$f(x)=x+\int_0^{\pi/2}\sin(x+y)f(y)dy$. Then $(a+b)$ is equal to : **[JEE(Main)Jan.2023]**

माना एक फलन $f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in IR$,

$f(x)=x+\int_0^{\pi/2}\sin(x+y)f(y)dy$ को संतुष्ट करता है। तब $(a+b)$ का मान है : **[JEE(Main)Jan.2023]**

(A) $-\pi(\pi+2)$ (B) $-2\pi(\pi+2)$ (C) $-2\pi(\pi-2)$ (D) $-\pi(\pi-2)$

Ans. B

Sol. $f(x)=x+\int_0^{\pi/2}(\sin x\cos y+\cos x\sin y)f(y)dy$

$f(x)=x+\int_0^{\pi/2}\left((\cos yf(y)dy)\sin x+(\sin yf(y)dy)\cos x\right)...(1)$

On comparing with

$f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in R$, then

$\Rightarrow\frac{a}{\pi^2-4}=\int_0^{\pi/2}\cos yf(y)dy$ ...(2)

$\Rightarrow\frac{b}{\pi^2-4}=\int_0^{\pi/2}\sin y\ f(y)dy$ ...(3)

Add (2) and (3)

$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f(y)dy$ ....(4)

$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f\left(\frac{\pi}{2}-y\right)dy$ ....(5)

Add (4) and (5)

$\frac{2(a+b)}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)\left(\frac{\pi}{2}+\frac{(a+b)}{\pi^2-4}(\sin y+\cos y)\right)dy$

$$= \pi + \frac{a+b}{\pi^2 - 4}\left(\frac{\pi}{2} + 1\right)$$

$(a + b) = -2\pi\,(\pi + 2)$

9. The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ is equal to : **[JEE(Main)Jan.2023]**

समाकलन $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ का मान बराबर है **[JEE(Main)Jan.2023]**

(A) $\pi \log_e 2$ (B) $\frac{1}{2}\log_e 2$ (C) $\frac{\pi}{4}\log_e 2$ (D) $\frac{\pi}{2}\log_e 2$

Ans. D

Sol. $I = \int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ .....(i)

Put $x = \frac{1}{t}$ $dx = -\frac{1}{t^2} dt$

$$I = -\int_{2}^{1/2} \frac{\tan^{-1}\frac{1}{t}}{\frac{1}{t}} \cdot \frac{1}{t^2} dt = -\int_{2}^{1/2} \frac{\tan^{-1}\frac{1}{t}}{t} dt$$

$$I = \int_{2}^{1/2} \frac{\cot^{-1} t}{t} dt = \int_{2}^{1/2} \frac{\cot^{-1} x}{x} dx \qquad .....(ii)$$

Add both equation

$$2I = \int_{1/2}^{2} \frac{\tan^{-1} x + \cot^{-1} x}{x} dx = \frac{\pi}{2}\int_{1/2}^{2} \frac{dx}{x} = \frac{\pi}{2}(\ell n\, 2)_{1/2}^{2}$$

$$= \frac{\pi}{2}\left(\ell n 2 - \ell n \frac{1}{2}\right) = \pi \ell n 2$$

$$I = \frac{\pi}{2}\ell n 2$$

10. The value of the integral $\int_{1}^{2}\left(\frac{t^4+1}{t^6+1}\right) dt$ is : **[JEE(Main)Jan.2023]**

समाकलन $\int_{1}^{2}\left(\frac{t^4+1}{t^6+1}\right) dt$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\tan^{-1}\frac{1}{2} - \frac{1}{3}\tan^{-1} 8 + \frac{\pi}{3}$ (B) $\tan^{-1}\frac{1}{2} + \frac{1}{3}\tan^{-1} 8 - \frac{\pi}{3}$

(C) $\tan^{-1}2-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$ (D) $\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

Ans. D

Sol. $=\int_1^2\frac{\left(t^4+1\right)}{\left(t^6+1\right)}\frac{\left(t^2+1\right)}{\left(t^2+1\right)}dt$

$=\int_1^2\frac{t^6+t^4+t^2+1}{\left(t^6+1\right)\left(t^2+1\right)}dt$

$=\int_1^2\frac{t^6+1}{\left(t^6+1\right)\left(t^2+1\right)}dt+\int_1^2\frac{t^4+t^2}{\left(t^6+1\right)\left(t^2+1\right)}dt$

$=\int_1^2\frac{1}{t^2+1}dt+\int_1^2\frac{t^2}{t^6+1}dt,$

$I=\left[\tan^{-1}t\right]_1^2+I_1$

$\tan^{-1}-\frac{\pi}{4}+I_1\ \tan^{-1}-\frac{\pi}{4}+I_1$

$I_1=\int_{11}^2\frac{t^2}{t^6+1}dt$

put $t^3=\theta\cdot 3t^2\,dt=d\theta$

$t=1$ then $\theta=1$

$t=2$ then $\theta=8$

$=\frac{1}{3}\int_1^8\frac{d\theta}{\theta^2+1}$

$=\frac{1}{3}\left[\tan^{-1}(\theta)\right]_1^8$

$=\frac{1}{3}\left[\tan^{-1}8-\frac{\pi}{4}\right]$

$I=\tan^{-1}2-\tan^{-1}(1)+\frac{1}{3}\tan^{-1}8-\frac{\pi}{12}$

$\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

11. If $[t]$ denotes the greatest integer $\le t$, then the value of $\frac{3(e-1)}{e}\int_1^2 x^2e^{[x]+\left[x^3\right]}dx$ is :

यदि [t] महत्तम पूर्णांक $\le t$, $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+[x^3]}dx$ का मान है : **[JEE(Main)Jan.2023]**

(A) $e^8 - 1$ (B) $e^7 - 1$ (C) $e^8 - e$ (D) $e^9 - 1$

Ans. C

Sol. $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+[x^3]}dx$

Let $k = \int_1^2 x^2 e^{1+[x^3]}dx$

$x^3 = t$

$3x^2\,dx = dt$

$k = \frac{1}{3}\int_1^8 e.e^{[t]}dt$

$\frac{e}{3}\left[e + e^2 + e^3 + ... + e^7\right] = \frac{e^2}{3}\left(\frac{e^7-1}{e-1}\right)$

$\frac{3(e-1)}{e}k = \frac{3(e-1)}{e}\frac{e^2}{3}\left(\frac{e^7-1}{e-1}\right) = e(e^7-1)$

$= e^8 - e$

12. $\lim_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+.....+\left(3-\frac{1}{n}\right)^2\right\}$ is equal to : **[JEE(Main)Jan.2023]**

$\lim_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+.....+\left(3-\frac{1}{n}\right)^2\right\}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 12 (B) 19 (C) $\frac{19}{3}$ (D) 0

Ans. B

Sol. $\lim_{n\to\infty}\frac{3}{n}\sum_{r=0}^{n-1}\left(2+\frac{r}{n}\right)^2$

$= 3\int_0^1 (2+x)^2 dx = 27 - 8 = 19$

13. The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ is equal to : **[JEE(Main)Jan.2023]**

$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ का मान बराबर है **[JEE(Main)Jan.2023]**

(A) $\frac{10}{3}-\sqrt{3}-\log_e\sqrt{3}$ (B) $-2+3\sqrt{3}+\log_e\sqrt{3}$

(C) $\frac{10}{3}-\sqrt{3}+\log_e\sqrt{3}$ (D) $\frac{7}{2}-\sqrt{3}-\log_e\sqrt{3}$

Ans. C

Sol. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2(1-\cos x)}{\sin x\left(1-\cos^2 x\right)}dx+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{3}{1+\cos x}dx$

$=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2}{\sin^3 x}dx-2\int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \cot x.\cos ec^2 x dx+\frac{3}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2\frac{x}{2}dx$

$=2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sqrt{1+\cot^2 x}.\cos ec^2 x dx-2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot x.\cos ec^2 x ds+3\left[\tan\frac{x}{2}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$

Let $\cot x = 1$

$=-2\int_{\frac{1}{\sqrt{3}}}^{0} \sqrt{1+t^2}dt+2\int_{\frac{1}{\sqrt{3}}}^{0} tdt+3\left[1-\frac{1}{\sqrt{3}}\right]$

$=2\left[\frac{t}{2}\sqrt{1+t^2}+\frac{1}{2}\log\left(t+\sqrt{1+t^2}\right)\right]_0^{\frac{1}{\sqrt{3}}}-\left[t^2\right]_0^{\frac{1}{\sqrt{3}}}+3-\sqrt{3}$

$=\frac{2}{3}+\log\sqrt{3}-\frac{1}{3}+3-\sqrt{3}=\frac{10}{3}-\sqrt{3}+\frac{1}{2}\log 3$

14. Let $\alpha\in(0,1)$ and $\beta=\log_e(1-\alpha)$. Let $P_n(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+.....+\frac{x^n}{n}, x\in(0,1)$. Then the integral $\int_0^{\alpha}\frac{t^{50}}{1-t}dt$ is equal to : **[JEE(Main)Jan.2023]**

माना $\alpha\in(0,1)$ तथा $\beta=\log_e(1-\alpha)$ है। माना $P_n(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+.....+\frac{x^n}{n}, x\in(0,1)$ है। तो समाकलन $\int_0^{\alpha}\frac{t^{50}}{1-t}dt$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $-\left(\beta+P_{50}(\alpha)\right)$ (B) $\beta+P_{50}(\alpha)$ (C) $\beta-P_{50}(\alpha)$ (D) $P_{50}(\alpha)-\beta$

Ans. A

Sol. $\int_0^{\alpha}\frac{t^{50}-1+1}{1-t}=-\int_0^{\alpha}\left(1+t+.....+t^{49}\right)+\int_0^{\alpha}\frac{1}{1-t}dt$

$$=-\left(\frac{\alpha^{50}}{50}+\frac{\alpha^{49}}{49}+.....+\frac{\alpha^{1}}{1}\right)+\left(\frac{\ln(1-f)}{-1}\right)_0^{\alpha}$$

$= -P_{50}(\alpha) - \ln(1-\alpha)$

$= \square\Pi_{50}(\alpha) - \beta$

15. Let $\alpha>0$. If $\int_0^{\alpha}\frac{x}{\sqrt{x+\alpha}-\sqrt{x}}dx=\frac{16+20\sqrt{2}}{15}$, then α is equal to : **[JEE(Main)Jan.2023]**

माना $\alpha>0$ है। यदि $\int_0^{\alpha}\frac{x}{\sqrt{x+\alpha}-\sqrt{x}}dx=\frac{16+20\sqrt{2}}{15}$ है, तो α बराबर है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{2}$ (B) 4 (C) $2\sqrt{2}$ (D) 2

Ans. D

Sol. After rationalising

$\int_0^{\alpha}\frac{x}{\alpha}(\sqrt{x+\alpha}+\sqrt{x})$

$\int_0^{\alpha}\frac{1}{\alpha}\left[(x+\alpha)^{3/2}-\alpha(x+\alpha)^{1/2}+x^{3/2}\right]$

$\frac{1}{\alpha}\left[\frac{2}{5}(x+\alpha)^{5/2}-\alpha\frac{2}{3}(x+\alpha)^{3/2}+\frac{2}{5}x^{5/2}\right]\Bigg|_0^{\alpha}$

$=\frac{1}{\alpha}\left(\frac{2}{5}(2\alpha)^{5/2}-\frac{2\alpha}{3}(2\alpha)^{3/2}+\frac{2}{5}\alpha^{5/2}-\frac{2}{5}\alpha^{5/2}+\frac{2}{3}\alpha^{5/2}\right.$

$=\frac{1}{\alpha}\left(\frac{2^{7/2}\alpha^{5/2}}{5}-\frac{2^{5/2}\alpha^{5/2}}{3}+\frac{2}{3}\alpha^{5/2}\right)$

$=\alpha^{3/2}\left(\frac{2^{7/2}}{5}-\frac{2^{5/2}}{3}+\frac{2}{3}\right)$

$=\frac{\alpha^{3/2}}{15}(24\sqrt{2}-20\sqrt{2}+10)=\frac{\alpha^{3/2}}{15}(4\sqrt{2}+10)$

Now, $\frac{\alpha^{\frac{3}{2}}}{15}\left(4\sqrt{2}+10\right)=\frac{16+20\sqrt{2}}{15}$

$\Rightarrow \alpha = 2$

16. If $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$, $x>0$, then $\phi'\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$, $x>0$ है, तो $\phi'\left(\frac{\pi}{4}\right)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{8}{6+\sqrt{\pi}}$ (B) $\frac{4}{6+\sqrt{\pi}}$ (C) $\frac{8}{\sqrt{\pi}}$ (D) $\frac{4}{6-\sqrt{\pi}}$

Ans. A

Sol. $\phi'(x)=\frac{1}{\sqrt{x}}\left[\left(4\sqrt{2}\sin x-3\phi'(x)\right)\cdot 1-0\right]-\frac{1}{2}x^{-3/2}$

$\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$

$\phi'\left(\frac{\pi}{4}\right)=\frac{2}{\sqrt{\pi}}\left[4-3\phi'\left(\frac{\pi}{4}\right)\right]+0$

$\left(1+\frac{6}{\sqrt{\pi}}\right)\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}}$

$\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}+6}$

17. $\lim_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+.....+\frac{1}{2n}\right]$ is equal to : **[JEE(Main)Jan.2023]**

$\lim_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+.....+\frac{1}{2n}\right]$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\log_e 2$ (B) $\log_e\left(\frac{2}{3}\right)$ (C) 0 (D) $\log_e\left(\frac{3}{2}\right)$

Ans. A

Sol. $\lim_{n\to\infty}\left(\frac{1}{1+n}+\frac{1}{2+n}+....+\frac{1}{2n}\right)=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{r+n}$

$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{\left(\frac{r}{n}+1\right)}\cdot\frac{1}{n}$

$\int_0^1\frac{dx}{1+x}=\left[\ln(1+x)\right]_0^1$

$\ln 2 - \ln 1 = \ln 2$

18. The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ is : **[JEE(Main)Jan.2023]**

समाकलन $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi^2}{6\sqrt{3}}$ (B) $\frac{\pi^2}{6}$ (C) $\frac{\pi^2}{3\sqrt{3}}$ (D) $\frac{\pi^2}{12\sqrt{3}}$

Ans. A

Sol. $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ (A)

$x \to -x$

$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{-x+\frac{\pi}{4}}{2-\cos 2x}dx$ (B)

(A) + (2)

$2I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\frac{\pi}{2}}{2-\cos 2x}dx$

$I = \frac{\pi}{4}\cdot 2\int_{0}^{\frac{\pi}{4}} \frac{dx}{2-\cos 2x}dx$

$I = \frac{\pi}{4}\cdot 2\int_{0}^{\frac{\pi}{4}} \frac{\left(1+\tan^2 x\right)dx}{2\left(1+\tan^2 x\right)-\left(1-\tan^2 x\right)}$

$I = \frac{\pi}{2}\int_{0}^{1} \frac{dt}{3t^2+1}$

$$\Rightarrow I = \frac{\pi}{2\sqrt{3}} \tan^{-1}\sqrt{3}$$

$$I = \frac{\pi^2}{6\sqrt{3}}$$

19. Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$. such that f(x) > 0 and $f(x) + \int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt = e$, $\forall x \in \left[0, \frac{\pi}{2}\right]$. Then $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\left[0, \frac{\pi}{2}\right]$ पर परिभाषित एक अवकलनीय फलन f है जिसके लिए f(x) > 0 और $f(x) + \int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt = e$, $\forall x \in \left[0, \frac{\pi}{2}\right]$ है। तब $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ बराबर है _____ **[JEE(Main)Jan.2023]**

Ans. 27

Sol. $t(x) = \int_0^x f(t)\sqrt{1-\ell nt(x)^2}\,dt = e : \nabla x \in \left[0, \frac{\pi}{2}\right]$

Differentiate on both side

$$f\cdot(x) + f(x)\sqrt{1-knf\left(xm^2\right.} = 0$$

$$\int \frac{f(x)dx}{f(x)\sqrt{1-tnf(x)m^2}} = -\int dx$$

Let $(n+\alpha) = U$

$$\frac{1}{f(x)}\cdot f'(x)dx = du$$

$$\int \frac{du}{\sqrt{1-u^2}} = -\int dx$$

$$\sin^{-1}\left(\ell n\left(f\left(x\right)\right)\right) = -x + c$$

put $x = 0$

$$\sin^{-1}\left(\ell n\left(f\left(0\right)\right)\right) = 0 + c$$

$$\sin^{-1}(1) = c$$

$$c = \frac{\pi}{2}$$

$$\sin^{-1}\left(\ell n\left(f\left(x\right)\right)\right) = -x + \frac{\pi}{2}$$

$$\ell n\left(f\left(x\right)\right)=\sin\left(\frac{\pi}{2}-x\right)=\cos x$$

$$f\left(x\right)=e^{\cos x}$$

$$f\left(\frac{\pi}{6}\right)=e^{\frac{\sqrt{3}}{2}}$$

$$\text{Now}\left(6\ell ne^{\frac{\sqrt{3}}{2}}\right)^2=\left(6\times\frac{\sqrt{3}}{2}\right)^2=27$$

20. The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{\left(\cos x\right)^{2023}}{\left(\sin x\right)^{2023}+\left(\cos x\right)^{2023}}dx$ is _____. **[JEE(Main)Jan.2023]**

$\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{\left(\cos x\right)^{2023}}{\left(\sin x\right)^{2023}+\left(\cos x\right)^{2023}}dx$ का मान है _____. **[JEE(Main)Jan.2023]**

Ans. 2

Sol. $I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ _____(1)

Using $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\sin x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ _____(2)

Adding (1) and (2)

$2I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}1dx$

$I=2$

21. The value of $12\int_0^3\left|x^2-3x+2\right|dx$ is _____.

$12\int_0^3\left|x^2-3x+2\right|dx$ का मान है _____

Ans. 22

Sol. $12\int_0^3\left|x^2-3x+2\right|dx$

$=12\int_0^3\left|\left(x-\frac{3}{2}\right)^2-\frac{1}{4}\right|dx$

If $x-\frac{3}{2}=t$

$dx = dt$

$$= 24\int_0^{3/2}\left|t^2-\frac{1}{4}\right|dt$$

$$= 24\left[-\int_0^{1/2}\left(t^2-\frac{1}{4}\right)dt+\int_{1/2}^{3/2}\left(t^2-\frac{1}{4}\right)dt\right]=22$$

22. If $\int_{\frac{1}{3}}^{3}|\log_e x|\,dx=\frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then m^2+n^2-5 is equal _____.

[JEE(Main)Jan.2023]

यदि $\int_{\frac{1}{3}}^{3}|\log_e x|\,dx=\frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$ है, जहाँ m तथा n असहभाज्य धन पूर्णांक है, तो m^2+n^2-5 बराबर है।

[JEE(Main)Jan.2023]

Ans. 20

Sol. $\int_{\frac{1}{3}}^{3}|\ell nx|\,dx=\int_{\frac{1}{3}}^{1}(-\ell nx)dx+\int_1^3(\ell nx)dx$

$$=-[x\ell nx-x]_{1/3}^1+[x\ell nx-x]_1^3$$

$$=-\left[-1-\left(\frac{1}{3}\ln\frac{1}{3}-\frac{1}{3}\right)\right]+[3\ln 3-3-(-1)]$$

$$=\left[-\frac{2}{3}-\frac{1}{3}\ln\frac{1}{3}\right]+[3\ln 3-2]$$

$$=-\frac{4}{3}+\frac{8}{3}\ln 3$$

$$=\frac{4}{3}(2\ln 3-1)$$

$$=\frac{4}{3}\left(\ell n\frac{9}{2}\right)$$

$\therefore m=4, n=3$

Now, $m^2+n^2-5=16+9-5=20$

23. $\lim_{x\to 0}\frac{48}{x^4}\int_0^x\frac{t^3}{t^6+1}dt$ is equal to _____. **[JEE(Main)Jan.2023]**

$\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x\dfrac{t^3}{t^6+1}dt$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 12

Sol. $\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x\dfrac{t^3}{t^6+1}dt$

$$=\lim_{x\to 0}\frac{48\int\limits_0^x\frac{t^3}{t^6+1}\cdot dt}{x^4}$$

$$=\lim_{x\to 0}\frac{48\times\frac{x^3}{x^6+1}}{4x^3}=\lim_{x\to 0}\frac{12}{x^6+1}=12$$

24. If $\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}=\dfrac{k\pi}{16}$, then k is equal to _____ .

[JEE(Main)Jan.2023]

यदि $\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}=\dfrac{k\pi}{16}$ है, तो k बराबर है।

[JEE(Main)Jan.2023]

Ans. 26

Sol. $I=\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}$

$$=2\int_0^{\pi/2}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx$$

$$=\int_0^{\pi/2}2+\cos 4x+\cos 2x+1+\cos 2x+2\cos 3x\left(\frac{\cos 3x+3\cos x}{4}\right)dx$$

$$=\int_0^{\pi/2}3dx+0+0+0+\int_0^{\pi/2}\frac{1}{2}\left(\cos^2 3x+3\cos x\cos 3x\right)dx$$

$$=\frac{3\pi}{2}+\frac{1}{4}\int_0^{\pi/2}\left(1+\cos 6x+3\left(\cos 4x+\cos 2x\right)\right)dx$$

$$= \frac{3\pi}{4} + \frac{1}{4}\cdot\frac{\pi}{2} = \frac{13\pi}{8} = \frac{k\pi}{16}$$

$\Rightarrow k = 26$

25. Let $I(x) = \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x + 1)^2}dx$. If $I(0) = 0$, then $I\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)April,2023]**

माना $I(x) = \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x + 1)^2}dx$ है। यदि $I(0) = 0$ है, तब $I\left(\frac{\pi}{4}\right)$ का मान है : **[JEE(Main)April,2023]**

(A) $\log_e \frac{(\pi+4)^2}{16} + \frac{\pi^2}{4(\pi+4)}$ (B) $\log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$

(C) $\log_e \frac{(\pi+4)^2}{32} + \frac{\pi^2}{4(\pi+4)}$ (D) $\log_e \frac{(\pi+4)^2}{16} - \frac{\pi^2}{4(\pi+4)}$

Ans. B

Sol. $\int x^2\left(\frac{x\sec^2 x + \tan x}{(x\tan x + 1)^2}\right)dx$

$$= \frac{x^2}{(x\tan x + 1)} + \int \frac{2x}{x\tan x + 1}dx$$

$$I = 2\int \frac{x}{x\tan x + 1}dx$$

$$= 2\int \frac{x\cos x}{x\sin x + \cos x}dx$$

Let $x\sin x + \cos x = t$

$(x\cos x + \sin x - \sin x)dx = dt$

$$= 2\int \frac{dt}{t} = 2\log t + c$$

$$= 2\log|x\sin x + \cos x| + c$$

$$\therefore \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{\left(x\tan + 1\right)^2}dx = \frac{-x^2}{x\tan x + 1} + 2\log\left|x\sin x + \cos x\right| + c$$

$I(0) = 0 \qquad \Rightarrow c = 0$

$$I\left(\frac{\pi}{4}\right) = \frac{-\pi}{\frac{\pi}{4}\times 1 + 1} + 2\log\left|\frac{1}{\sqrt{2}}\left(\frac{\pi}{4}\right) + 1\right|$$

$$= \log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$$

26. Let $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx, x>0$. If $\lim\limits_{x\to\infty} I(x)=0$, then $I(1)$ is equal to : **[JEE(Main)April,2023]**

माना $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx, x>0$ है। यदि $\lim\limits_{x\to\infty} I(x)=0$ है, तो $I(1)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{e+2}{e+1}-\log_e(e+1)$ (B) $\frac{e+1}{e+2}+\log_e(e+1)$

(C) $\frac{e+1}{e+2}-\log_e(e+1)$ (D) $\frac{e+2}{e+1}+\log_e(e+1)$

Ans. A

Sol. $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx$

$1+xe^x=t$

$\left(xe^x+e^x\right)dx=dt$

$(x+1)dx=\frac{1}{e^x}dt$

$\therefore\int\frac{dt}{xe^x\cdot t^2}=\int\frac{dt}{(t-1)t^2}=\int\frac{dt}{t(t-1)\cdot t}\Rightarrow\int\frac{t-(t-1)}{t(t-1)t}dt$

$=\int\frac{dt}{t(t-1)}-\int\frac{dt}{t^2}=\int\frac{t-(t-1)}{t(t-1)}dt+\frac{1}{t}+c$

$=\ln|t-1|-\ln|t|+\frac{1}{t}+c$

$\Rightarrow\ln\left|xe^x\right|-\ln\left|1+xe^x\right|+\frac{1}{1+xe^x}+c$

$I(x)=\ln\left|\frac{xe^x}{1+xe^x}\right|+\frac{1}{1+xe^x}+c$

$\lim\limits_{x\to\infty} I(x)=c=0$

$\therefore I(1)=\ln\left|\frac{e}{1+e}\right|+\frac{1}{1+e}$

$=\ln e-\ln(1+e)+\frac{1}{1+e}$

$=\frac{e+2}{e+1}-\ln(1+e)$

27. The integral $\int\left(\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right)\log_2 x dx$ is equal to : **[JEE(Main)April,2023]**

समाकलन $\int\left(\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right)\log_2 x dx$ बराबर है : **[JEE(Main)April,2023]**

(A) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{x}{2}\right)+C$ (B) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{2}{x}\right)+C$

(C) $\left(\frac{x}{2}\right)^x-\left(\frac{2}{x}\right)^x+C$ (D) $\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x+C$

Ans. B **Bonus**

Sol. $\int\left(x^x 2^{-x}+2^x x^{-x}\right)\log_2 x dx$

$\int\left(e^{x\ln x}\cdot e^{-x\ln 2}+e^{x\ln 2}\cdot e^{-x\ln x}\right)dx$

$\int\left(e^{x\ln x-x\ln 2}+e^{x\ln 2-x\ln x}\right)\frac{\ln x}{\ln 2}dx$

let $x\ln x-x\ln 2=t$

$(\ln x+1-\ln 2)dx=dt$

28. If $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ and $I(0)=1$, then $I\left(\frac{\pi}{3}\right)$ is equal to : **[JEE(Main)April,2023]**

यदि $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ तथा $I(0)=1$ हैं, तो $I\left(\frac{\pi}{3}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) $e^{\frac{3}{4}}$ (B) $-\frac{1}{2}e^{\frac{3}{4}}$ (C) $-e^{\frac{3}{4}}$ (D) $\frac{1}{2}e^{\frac{3}{4}}$

Ans. D

Sol. $\because I=\int e^{f(x)}\left(g'(x)+f'(x)\cdot g(x)\right)dx=e^{f(x)}\cdot g(x)$

$\therefore I=e^{\sin^2 x}\cdot\cos x+c$

but $I(0)=1\Rightarrow c=0$

$I=e^{\sin^2 x}\cos x$

$I\left(\frac{\pi}{3}\right)=\frac{1}{2}e^{\frac{3}{4}}$

29. For $\alpha, \beta, \gamma, \delta\in\mathbb{N}$, if $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$, where $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ and C is constant of integration, then $\alpha+2\beta+3\gamma-4\delta$ is equal to : **[JEE(Main)April,2023]**

$\alpha, \beta, \gamma, \delta \in \mathbb{N}$ के लिए, यदि $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$ है, जहाँ $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ तथा C समाकलन अचर है, तो $\alpha + 2\beta + 3\gamma - 4\delta$ बराबर है : **[JEE(Main)April,2023]**

(A) –4 (B) 1 (C) 4 (D) –8

Ans. C

Sol. $\text{Let}\left(\frac{x}{e}\right)^{2x}=t$

$2x(\ln x-1)=\ln t$

$\left(2(\ln x-1)+2x\times\frac{1}{x}\right)dx=\frac{1}{t}dt$

$\ln x\,dx=\frac{1}{2t}dt$

$I=\int\left(t+\frac{1}{t}\right)\times\frac{1}{2t}dt=\frac{1}{2}\int\left(1+\frac{1}{t^2}\right)dt$

$=\frac{1}{2}\left(t-\frac{1}{t}\right)+C$

$=\frac{1}{2}\left(\frac{x}{e}\right)^{2x}-\frac{1}{2}\left(\frac{e}{x}\right)^{2x}+C$

$\Rightarrow \alpha=2,\beta=2,\gamma=2,\delta=2$

$\therefore\quad \alpha+2\beta+3\gamma-4\delta=2+4+6-8=4$

30. Let $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ and $I(9)=12+7\log_e 7$. If $I(1)=\alpha+7\log_e(1+2\sqrt{2})$, α^4 is equal to _____.

[JEE(Main)April,2023]

माना $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ तथा $I(9)=12+7\log_e 7$ हैं। यदि $I(1)=\alpha+7\log_e(1+2\sqrt{2})$ है, तो α^4 बराबर है _____

[JEE(Main)April,2023]

Ans. 64

Sol. $I(x)=\int\sqrt{\frac{x+7}{x}}dx$

$=\int\frac{\sqrt{\left(\sqrt{x}\right)^2+\left(\sqrt{7}\right)^2}}{\sqrt{x}}dx$

Let $\sqrt{x} = t \therefore \frac{1}{2\sqrt{x}}dx = dt$

$= 2\int \sqrt{t^2 + (\sqrt{7})^2}\,dt = 2\left(\frac{t}{2}\sqrt{t^2+7} + \frac{7}{2}\ln\left(t + \sqrt{t^2+7}\right)\right) + c$

$I(x) = \sqrt{x}\sqrt{x+7} + 7\ln(\sqrt{x} + \sqrt{x+7}) + c$

$I(9) = 12 + 7\ln 7 = 12 + 7\ln(3+4) + c$

$\Rightarrow c = 0$

$\therefore I(x) = \sqrt{x}\sqrt{x+7} + 7\ln(\sqrt{x} + \sqrt{x+7})$

at $x = 1$

$I(1) = \sqrt{8} + 7\ell n(1+\sqrt{8}) = 7\ell n(1+2\sqrt{2}) + \alpha$

$\alpha = \sqrt{8}$

$\alpha^4 = 64$

31. Let $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$. If $f(0) = 0$ and $f(1) = \frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$, then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)April,2023]**

यदि $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$, $f(0) = 0$ तथा $f(1) = \frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$ हो, तो $\alpha^2 + \beta^2$ का मान gksxk_____ **[JEE(Main)April,2023]**

Ans. 28

Sol. $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$

Let $x = \frac{1}{t}$

$dx = -\frac{1}{t^2}dt$

$= \int \frac{-\frac{1}{t^2}dt}{\left(3+\frac{4}{t^2}\right)\sqrt{4-\frac{3}{t^2}}}$

$= \int \frac{-tdt}{\left(3t^2+4\right)\sqrt{4t^2-3}}$

Now, let $4t^2 - 3 = p^2$

$4t\,dt = p\,dp$

$$= \int \frac{-\frac{p}{4}dp}{\left(\frac{3p^2+25}{4}\right)p} = -\int \frac{dp}{3p^2+25}$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}p}{5}\right) + C$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4t^2-3}}{5}\right) + C$$

$$= -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{\frac{4}{x^2}-3}}{5}\right) + C$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4-3x^2}}{5|x|}\right) + C$$

$x = 0, y = 0$

$$0 = -\frac{1}{5\sqrt{3}} \tan^{-1}(\infty) + C$$

$$C = \frac{1}{5\sqrt{3}} \times \frac{\pi}{2}$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\frac{\sqrt{3}\sqrt{4-3x^2}}{5x} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$f(1) = -\frac{1}{5\sqrt{3}} \tan^{-1}\frac{\sqrt{3}}{5} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$= \frac{1}{5\sqrt{3}}\left(\frac{\pi}{2} - \tan^{-1}\frac{\sqrt{3}}{5}\right)$$

$$= \frac{1}{5\sqrt{3}} \cot^{-1}\frac{\sqrt{3}}{5}$$

$$= \frac{1}{5\sqrt{3}} \tan^{-1}\frac{5}{\sqrt{3}}$$

Now, $\alpha = 5,\ \beta = \sqrt{3}$

$\alpha^2 + \beta^2 = 25 + 3 = 28$

32. Let $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$. Then $18\int_1^2 f(x)dx$ is equal to : **[JEE(Main)April,2023]**

माना $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$ है। तब $18\int_1^2 f(x)dx$ का मान है : **[JEE(Main)April,2023]**

(A) $5\log_e 2-3$ (B) $5\log_e 2+3$ (C) $10\log_e 2-6$ (D) $10\log_e 2+6$

Ans. C

Sol. $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3$

$x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=x+3$

$(1)\times 5-(2)\times 4$

$\Rightarrow f(x)=\frac{5}{9x}-\frac{4}{9}x+\frac{1}{3}$

$\Rightarrow 18\int_1^2 f(x)dx=18\left(\frac{5}{9}\ln 2-\frac{4}{9}\times\frac{3}{2}+\frac{1}{3}\right)$

$=10\ln 2-6$

33. Let f(x) be a function satisfying $f(x)+f(\pi-x)=\pi^2, \forall x\in\mathbb{R}$. Then $\int_0^\pi f(x)\sin x dx$ is equal to : **[JEE(Main)April,2023]**

माना एक फलन f(x) के लिए $f(x)+f(\pi-x)=\pi^2, \forall x\in\mathbb{R}$ है। तो $\int_0^\pi f(x)\sin x dx$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{\pi^2}{2}$ (B) π^2 (C) $\frac{\pi^2}{4}$ (D) $2\pi^2$

Ans. B

Sol. $I=\int_0^\pi f(x)\sin x dx$

$I=\int_0^\pi f(\pi-x)\sin x dx$

$2I=\int_0^\pi \sin x(f(x)+f(\pi-x))dx$

$2I=\pi^2\int_0^\pi \sin x dx$

$2I=2\pi^2\int_0^{\frac{\pi}{2}}\sin x dx$

$I = \pi^2$

34. Let f be a differentiable function such that $x^2 f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$. Then 18f(3) is equal to :

[JEE(Main)April,2023]

माना एक अवकलनीय फलन f के लिए $x^2 f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$ हैं। तो 18f(3) बराबर है :

[JEE(Main)April,2023]

(A) 180 (B) 210 (C) 160 (D) 150

Ans. C

Sol. Differentiate the given equation

$\Rightarrow 2xf(x) + x^2 f'(x) - 1 = 4xf(x)$

$\Rightarrow x^2 \frac{dy}{dx} - 2xy = 1$

$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right) y = \frac{1}{x^2}$

$I.F. = e^{\int -\frac{2}{x} \ell nx} = \frac{1}{x^2}$

$\therefore y\left(\frac{1}{x^2}\right) = \int \frac{1}{x^4} dx$

$\Rightarrow \frac{y}{x^2} = \frac{-1}{3x^3} + c$

$\Rightarrow y = -\frac{1}{3x^3} + c$

$\Rightarrow y = -\frac{1}{3x} + cx^2$

$\because f(1) = \frac{2}{3} = -\frac{1}{3} + c \Rightarrow c = 1$

$f(x) = -\frac{1}{3x} + x^2$

$18f(3) = 160$

35. Let f be a continuous function satisfying $\int_0^{t^2} \left(f(x) + x^2\right) dx = \frac{4}{3} t^3, \forall t > 0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

[JEE(Main)April,2023]

माना f एक सतत फलन है तथा $\int_0^{t^2} \left(f(x) + x^2\right) dx = \frac{4}{3} t^3, \forall t > 0$ है। तो $f\left(\frac{\pi^2}{4}\right)$ बराबर है :

(A) $\left(1-\pi\left(1+\frac{\pi^3}{16}\right)\right.$ (B) $-\pi\left(1-\frac{\pi^2}{16}\right)$ (C) $\pi^2\left(1-\frac{\pi^3}{16}\right)$ (D) $\pi\left(1-\frac{\pi^3}{16}\right)$

Ans. D

Sol. $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3, \forall t>0$

$\left(f\left(t^2\right)+t^4\right)=2t$

$f\left(t^2\right)=2t-t^4$

$t=\frac{\pi}{2}\Rightarrow f\left(\frac{\pi^2}{4}\right)=\frac{2\pi}{2}-\frac{\pi^4}{16}$

$=\pi-\frac{\pi^4}{16}=\pi\left(1-\frac{\pi^3}{16}\right)$

36. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ is equal to : **[JEE(Main)April,2023]**

निम्न $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ का मान ज्ञात कीजिए : **[JEE(Main)April,2023]**

(A) $\log_e\left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$ (B) $\log_e\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(C) $\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$ (D) $\log_e\left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

Ans. B

Sol. $I=\int_{-\log_e 2}^{\log_e 2} e^x\left[\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right]dx$

Put $e^x=t$

$e^x dx=dt$

$I=\int_{\frac{1}{2}}^{2} 1\times\log_e\left[t+\sqrt{1+t^2}\right]dt$

$=\left[t\ln\left(\sqrt{t^2+1}+t\right)\right]_{\frac{1}{2}}^{2}-\int_{1/2}^{2}\frac{t}{\sqrt{t^2+1}}dt$

$=\left[t\ln\sqrt{t^2+1}+t-\sqrt{t^2+1}\right]_{\frac{1}{2}}^{2}$

$$=\left[2\ln\sqrt{5}+2-\sqrt{5}\right]-\left[\frac{1}{2}\ln\left(\sqrt{\frac{5}{2}}+\frac{1}{2}\right)-\sqrt{\frac{5}{2}}\right]$$

$$=\ln\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$$

37. If $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$, then the value of α is : **[JEE(Main)April,2023]**

यदि $f:\mathbb{R}\to\mathbb{R}$ एक सतत फलन है तथा $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$ है, तो α का मान है : **[JEE(Main)April,2023]**

(A) $-\sqrt{3}$ (B) $\sqrt{2}$ (C) $-\sqrt{2}$ (D) $\sqrt{3}$

Ans. C

Sol. $I=\int_0^{\frac{\pi}{4}} f(\sin 2x)\sin x\,dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx$

$$+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

Apply king in first part and put $x-\frac{\pi}{4}=t$ in second part.

$$I=\int_0^{\frac{\pi}{4}} f(\cos 2x)\sin\left(\frac{\pi}{4}-x\right)dx+\int_0^{\frac{\pi}{4}} f(\cos 2t)\sin\left(\frac{\pi}{4}+t\right)dt$$

$$+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

$$I=\int_0^{\frac{\pi}{4}} f(\cos 2x)\left[2\sin\frac{\pi}{4}\cdot\cos x+\alpha\cos x\right]dx=0$$

$$I=(\alpha+\sqrt{2})\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

$$\therefore\alpha=-\sqrt{2}$$

38. Let the function $f:[0,2]\to\mathbb{R}$ be defined as $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\}}, & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ where [t] denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is : **[JEE(Main)April,2023]**

माना $f:[0,2]\to\mathbb{R}$, $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\},} & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ द्वारा परिभाषित है, जहाँ [t] का महत्तम पूर्णांक ≤ t है। तो समाकलन $\int_0^2 xf(x)dx$ का मान है : **[JEE(Main)April,2023]**

(A) $2e-1$ (B) $1+\frac{3e}{2}$ (C) $2e-\frac{1}{2}$ (D) $(e-1)\left(e^2+\frac{1}{2}\right)$

Ans. C

Sol. $f(x)=\begin{cases}e^{x^2} & x\in(0,1)\\ e & x\in[1,2)\end{cases}$

Now $\int_0^2 xf(x)dx=\int_0^1 xe^{x^2}dx+\int_1^2 x\cdot edx=2e-\frac{1}{2}$

39. Among

(S1) : $\lim\limits_{n\to\infty}\frac{1}{n^2}(2+4+6+\ldots\ldots+2n)=1$

(S2) : $\lim\limits_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right)=\frac{1}{16}$ **[JEE(Main)April,2023]**

(A) Only (S2) is true (B) Both (S1) and (S2) are true

(C) Only (S1) is true (D) Both (S1) and (S2) are false

(S1) : $\lim\limits_{n\to\infty}\frac{1}{n^2}(2+4+6+\ldots\ldots+2n)=1$

(S2) : $\lim\limits_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right)=\frac{1}{16}$ में से : **[JEE(Main)April,2023]**

(A) केवल (S2) सत्य है (B) दोनों (S1) तथा (S2) सत्य हैं

(C) केवल (S1) सत्य है (D) दोनों (S1) तथा (S2) असत्य हैं

Ans. B

Sol. $S_1:\lim\limits_{n\to\infty}\frac{n(n+1)}{n^2}=1\Rightarrow$ True

$S_2:\lim\limits_{n\to\infty}n^{\frac{1}{16}}\left(\sum r^{15}\right)=\lim\limits_{n\to\infty}\frac{1}{n}\sum\left(\frac{r}{n}\right)^{15}$

$=\int_0^1 x^{15}dx=\frac{1}{16}\Rightarrow$ True

40. $\int_0^\infty\frac{6}{e^{3x}+6e^{2x}+11e^x+6}dx=$ **[JEE(Main)April,2023]**

(A) $\log_e\left(\frac{256}{81}\right)$ (B) $\log_e\left(\frac{512}{81}\right)$ (C) $\log_e\left(\frac{32}{27}\right)$ (D) $\log_e\left(\frac{64}{27}\right)$

Ans. C

Sol. Let $e^x = t \Rightarrow e^x dx = dt$

$$I = \int_1^\infty \frac{6}{t(t+1)(t+2)(t+3)}dt$$

$$\Rightarrow I = 6\int_0^\infty \left\{\frac{A}{t} + \frac{B}{t+1} + \frac{C}{t+2} + \frac{D}{t+3}\right\}dt$$

$$\Rightarrow I = 6\int_1^\infty \left(\frac{1}{6t} + \frac{1}{(-2)(t+1)} + \frac{1}{2(t+2)} - \frac{1}{6(t+3)}\right)dt$$

$$\Rightarrow I = \left[\ell nt - 3\ell n(t+1) + 3\ell n(t+2) - \ell n(t+3)\right]_1^\infty$$

$$\Rightarrow I = \left[\ell n \frac{t(t+2)^3}{(t+1)^3(t+3)}\right]_1^\infty$$

$$\Rightarrow I = 0 - \ell n\frac{3^3}{2^3.4} = \ell n\frac{32}{27}$$

41. The value of $\frac{e^{-\frac{\pi}{4}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x + \tan^{51}x\right)dx}$ is : **[JEE(Main)April,2023]**

$\frac{e^{-\frac{\pi}{4}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x + \tan^{51}x\right)dx}$ का मान है : **[JEE(Main)April,2023]**

(A) 51 (B) 49 (C) 25 (D) 50

Ans. D

Sol. Let $I_1 = e^{\frac{-\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx$

$$I_2 = \int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49} x + \tan^{51} x\right)dx$$

$$= \int_0^{\frac{\pi}{4}} e^{-x} \tan^{49} x\left(\sec^2 x\right)dx$$

$$= \left|e^{-x}\frac{\tan^{50} x}{50}\right|_0^{\frac{\pi}{4}} + \frac{1}{50}\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50} x dx$$

$$= \frac{e^{-\frac{\pi}{4}}}{50} + \frac{1}{50}\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50} x dx = \frac{I_1}{50}$$

then $\frac{I_1}{I_2} = 50$

42. If $\int_0^1 \frac{1}{\left(5+2x-2x^2\right)\left(1+e^{(2-4x)}\right)}dx = \frac{1}{\alpha}\log_e\left(\frac{\alpha+1}{\beta}\right)$, $\alpha, \beta, > 0$, then $\alpha^4 - \beta^4$ is equal to :

[JEE(Main)April,2023]

यदि $\int_0^1 \frac{1}{\left(5+2x-2x^2\right)\left(1+e^{(2-4x)}\right)}dx = \frac{1}{\alpha}\log_e\left(\frac{\alpha+1}{\beta}\right)$, $\alpha, \beta, > 0$ हो, तो $\alpha^4 - \beta^4$ का मान होगा :

[JEE(Main)April,2023]

(A) 0 (B) 21 (C) –21 (D) 19

Ans. B

Sol. $$I = \int_0^1 \frac{dx}{\left(5+2x-2x^2\right)\left(1+e^{2-4x}\right)} \quad ...(i)$$

$x \to 1 - x$

$$I = \int_0^1 \frac{e^{2-4x}dx}{\left(5+2x-2x^2\right)\left(1+e^{2-4x}\right)} \quad ...(ii)$$

Add (i) and (ii)

$$2I = \int_0^1 \frac{dx}{5+2x-2x^2} = \int_0^1 \frac{dx}{2\left(\frac{11}{4} - \left(x-\frac{1}{2}\right)^2\right)}$$

$$I=\frac{1}{\sqrt{11}}\ln\left(\frac{\sqrt{11}+1}{\sqrt{10}}\right) \quad \begin{matrix}\alpha=\sqrt{11}\\ \beta=\sqrt{10}\end{matrix}$$

$$\alpha^4-\beta^4=121-100=21$$

43. Let $f(x)=\dfrac{x}{\left(1+x^n\right)^{\frac{1}{n}}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2.$

If $f^n(x)=\left(\text{fofof.....upto n times}\right)(x)$, then $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$ is equal to _____.

[JEE(Main)April,2023]

माना $f(x)=\dfrac{x}{\left(1+x^n\right)^{\frac{1}{n}}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2$ है।

; fn $f^n(x)=\left(\text{fofof.....upto n times}\right)(x)$ है तो $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$ बराबर है _____

[JEE(Main)April,2023]

Ans. 0

Sol. $\text{Let } f(x)=\dfrac{x}{1+x^n\frac{1}{n}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2.$

If $f^n(x)=n\ (\text{fofof....upto n times})\ (x)$

then $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$

$$f(f(x))=\frac{x}{\left(1+2x^n\right)^{1/n}}$$

$$f(f(f(x)))=\frac{x}{\left(1+3x^n\right)^{1/n}}$$

$$\text{Similarly } f^n(x)=\frac{x}{\left(1+n\cdot x^n\right)^{1/n}}$$

$$\text{Now } \lim_{n\to\infty}\int\frac{x^{n-2}\cdot x dx}{\left(1+n\cdot x^n\right)^{1/n}}=\lim_{n\to\infty}\int\frac{x^{n-1}\cdot dx}{\left(1+n\cdot x^n\right)^{1/n}}$$

$$\text{Now } 1+nx^n=t$$

$$n^2\cdot x^{n-1}dx=dt$$

$$x^{n-1}dx=\frac{dt}{n^2}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\int_1^{1+n}\frac{dt}{t^{1/n}}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\left[\frac{t^{1-\frac{1}{n}}}{1-\frac{1}{n}}\right]_1^{1+n}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n(n-1)}\left((1+n)^{\frac{n-1}{n}}-1\right) \text{ Now let } n=\frac{1}{h}$$

$$\Rightarrow \lim_{h\to 0}\frac{\left(1+\frac{1}{h}\right)^{1-h}-1}{\frac{1}{h}\frac{(1-h)}{h}}$$

Using series expansion

$\Rightarrow 0$

44. Let [t] denote the greatest integer $\le t$. Then $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\text{cosec}\,x]-5[\cot x])dx$ is equal to _____.

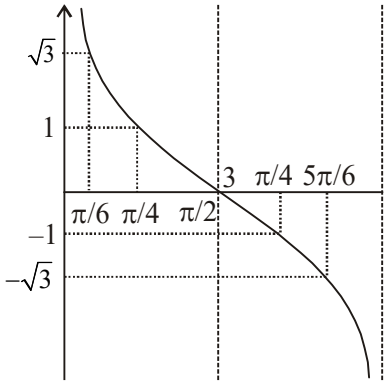
[JEE(Main)April,2023]

माना [t] महत्तम पूर्णांक $\le t$ है। तो $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\text{cosec}\,x]-5[\cot x])dx$ बराबर है _____

[JEE(Main)April,2023]

Ans. 14

Sol.



$$I=\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}\left(8[\text{cosec}\,x]-5[\cot x]\right)dx$$

For $x\in(\pi/6,5\pi/6)$

$\text{cosec}\,x\in[1,2)$

$[\text{cosec}\,x]=1$

$$I=\frac{2}{\pi}\left[\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}8(1)dx-\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}5(1)dx-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}5(0)dx-\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}5(-1)dx-\int_{\frac{3\pi}{4}}^{\frac{5\pi}{6}}5(-2)dx\right]=14$$

45. Let [t] denote the greatest integer function. If $\int_0^{2.4}\left[x^2\right]dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}+\delta\sqrt{5}$, then $\alpha+\beta+\gamma+\delta$ is equal to _____. **[JEE(Main)April,2023]**

माना [t] महत्तम पूर्णांक फलन है। यदि $\int_0^{2.4}\left[x^2\right]dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}+\delta\sqrt{5}$ है, तो $\alpha+\beta+\gamma+\delta$ बराबर है : **[JEE(Main)April,2023]**

Ans. 6

Sol. $\int_0^{2.4}\left[x^2\right]dx=\int_0^1 0dx+\int_1^{\sqrt{2}}1dx+\int_{\sqrt{2}}^{\sqrt{3}}2dx$

$+\quad\int_{\sqrt{3}}^{2}3dx+\int_2^{\sqrt{5}}4dx+\int_{\sqrt{5}}^{2.4}5dx$

$=0+(\sqrt{2}-1)+2(\sqrt{3}-\sqrt{2})+3(2-\sqrt{3})+4(\sqrt{5}-2)+5\left(2.4-\sqrt{5}\right)$

$=9-\sqrt{2}-\sqrt{3}-\sqrt{5}$

$\alpha=9,\beta=-1,\gamma=-1,\delta=-1$

46. For m, n > 0, let $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$. If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$, then p is equal to _____. **[JEE(Main)April,2023]**

m, n > 0 के लिए माना $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$ है। यदि $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ तो p बराबर _____ है। **[JEE(Main)April,2023]**

Ans. 32

Sol. $\alpha(m,n)=\int_0^2 t^m(1+3t)^n dt$

If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ then P

$=11\int_0^2\frac{t^{10}}{II}\frac{(1+3t)^6}{I}+10\int_0^2 t^{11}(1+3t)^5dt$

$=11\left[(1+3t)^6\cdot\frac{t^{11}}{11}-\int 6(1+3t)^5\cdot 3\frac{t^{11}}{11}\right]_0^2+18\int_0^2 t^{11}(1+3t)^5dt$

$=\left(t^{11}(1+3t)^6\right)_0^2$

$=2^{11}(7)^6$

$=2^5(14)^6$

$= 32(14)^6$

47. If $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = \frac{k}{3000}$, then k is equal to _____. **[JEE(Main)April,2023]**

यदि $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = \frac{k}{3000}$ है, तो k बराबर है _____ **[JEE(Main)April,2023]**

Ans. 575

Sol. $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = 2\int_{0}^{0.15} \left|100x^2 - 1\right| dx$

$= 2\int_{0}^{0.1} \left(-100x^2 + 1\right) dx + 2\int_{0.1}^{0.15} \left(100x^2 - 1\right) dx$

$= 2 \cdot \left(\frac{-100x^3}{3} + x\right)_0^{0.1} + 2\left(\frac{100x^3}{3} - x\right)_{0.1}^{0.15}$

$= 4\left(-\frac{1}{30} + \frac{1}{10}\right) + 2\left(\frac{9}{80} - \frac{3}{20}\right)$

$= \frac{8}{30} - \frac{3}{40} = \frac{575}{3000} \Rightarrow k = 575$

48. Let for $x \in \mathbb{R}, S_0(x) = x, S_k(x) = C_k x + k\int_0^x S_{k-1}(t)dt$, where $C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x)dx, k = 1, 2, 3, \ldots$ Then $S_2(3) + 6C_3$ is equal to _____. **[JEE(Main)April,2023]**

माना $x \in \mathbb{R}, S_0(x) = x, S_k(x) = C_k x + k\int_0^x S_{k-1}(t)dt$ है, जहाँ $C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x)dx, k = 1, 2, 3, \ldots$ हैं। तो $S_2(3) + 6C_3$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 18

Sol. $C_0 = 1$

$C_1 = 1 - \int_0^1 S_0(x)dx = 1 - \int_0^1 x dx = 1 - \frac{1}{2} = \frac{1}{2}$

$S_1(x) = C_1 x + \int_0^x t dt$

$= \frac{1}{2}x + \frac{x^2}{2} = \frac{x + x^2}{2}$

$C_2 = 1 - \int_0^1 \left(\frac{x + x^2}{2}\right) dx = 1 - \frac{1}{2}\left(\frac{1}{2} + \frac{1}{3}\right) = 1 - \frac{5}{12} = \frac{7}{12}$

$$S_2(x) = C_2 x + 2\int_0^x \left(\frac{t+t^2}{2}\right)dt = \frac{7}{12}x + \left(\frac{x^2}{2} + \frac{x^3}{3}\right) = \frac{7x+6x^2+4x^3}{12}$$

$$S_2(3) = \frac{21+54+108}{12} = \frac{183}{12} = \frac{61}{4}$$

$$C_3 = 1 - \int_0^1 \left(\frac{7x+6x^2+4x^3}{12}\right)dx$$

$$= 1 - \frac{1}{12}\left[\frac{7}{2} + \frac{6}{3} + \frac{4}{4}\right]$$

$$= 1 - \frac{1}{12}\left(\frac{7}{2} + 3\right) = \frac{24-7-6}{24} = \frac{11}{24}$$

Now, $S_2(3) + 6C_3 = \frac{61}{4} + \frac{11}{4} = \frac{72}{4} = 18$

49. Let $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^{n} \sin^{k-1} x\right)\left(\sum_{k=1}^{n} (2k-1)\sin^{k-1} x\right)\cos x dx, n \in \mathbb{N}$. Then $f_{21} - f_{20}$ is equal to _____.

[JEE(Main)April,2023]

माना $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^{n} \sin^{k-1} x\right)\left(\sum_{k=1}^{n} (2k-1)\sin^{k-1} x\right)\cos x dx, n \in \mathbb{N}$ है। तो $f_{21} - f_{20}$ बराबर है _____

[JEE(Main)April,2023]

Ans. 41

Sol. $f_n(x) = \int_0^{\frac{\pi}{2}} \left(1 + \sin x + \sin^2 x + \sin^3 x + ... + \sin^{n-1}(x)\right)\left(1 + 3\sin x + 5\sin^2 x + .. + (2n-1)\right)\sin^{n-1} x.\cos x dx$

Multiply & divide by $\sqrt{\sin x}$

$$\int_0^{\frac{\pi}{2}} \left((\sin x)^{\frac{1}{2}} + (\sin x)^{\frac{3}{2}} + (\sin x)^{\frac{5}{2}} + (\sin x)^{\frac{7}{2}} + ...(\sin x)^{\frac{2n-1}{2}}\right)$$

$$\left(1 + 3\sin x + 5\sin^2 x + ... + (2n-1)\sin^{n-1}(x)\right)\frac{\cos x}{\sqrt{\sin x}} dx$$

Put $(\sin x)^{1/2} + (\sin x)^{3/2} + (\sin x)^{5/2} + ... (\sin x)^{n-1/2} = t$

$$\frac{1}{2}\frac{\left(1 + 3\sin x + 5\sin^2 x + ...(2n-1)\sin^{n-1} x\right)}{\sqrt{\sin x}}\cos x dx = dt$$

$$f_n = 2\int_0^n t dt$$

$f_n = n^2$

$f_{21} - f_{20} = (21)^2 - (20)^2$

$= 441 - 400$

$= 41$

50. The integral $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to : **[JEE(Main)Jan.2024]**

समाकलन $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + C$ (B) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + C$

(C) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + C$ (D) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^3 + C$

Ans. A

Sol. $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$

$\tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$

$\frac{\left(3x^2 - \frac{3}{x^4}\right)dx}{1 + \left(x^3 + \frac{1}{x^3}\right)^2} = dt$

$\frac{3\left(x^6 - 1\right)dx}{x^4\left(1 + x^6 + \frac{1}{x^6} + 2\right)} = dt$

$\frac{3\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)} = dt$

$= \frac{1}{3}\int \frac{dt}{t}$

$$= \frac{1}{3}\ell n\left|t\right| + c$$

$$\log_e \left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|^{1/3} + c$$

51. For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\cos ec\, x + \sin x}{\cos ecx \sec x + \tan x \sin^2 x} dx$, and $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to :

[JEE(Main)Jan.2024]

$x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ के लिए यदि $y(x) = \int \frac{\cos ec\, x + \sin x}{\cos ecx \sec x + \tan x \sin^2 x} dx$ है तथा $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} y(x) = 0$ है, तो $y\left(\frac{\pi}{4}\right)$ बराबर है :

[JEE(Main)Jan.2024]

(A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$ (C) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Ans. B

Sol. $$\int \left(\frac{1 + \sin^2 x}{\frac{1}{\cos x} + \frac{\sin^4 x}{\cos x}}\right) dx$$

$$= \int \frac{\left(1 + \sin^2 x\right)\cos x}{1 + \sin^4 x} dx$$

$\sin x = t, \qquad \cos x\, dx = dt$

$$\int \frac{1 + t^2}{1 + t^4} dt = \int \left(\frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}}\right) dt$$

$$= \int \frac{du}{u^2 + 2} \qquad t - \frac{1}{t} = u$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}} + c \qquad \left(1 + \frac{1}{t^2}\right)dt = dx$$

$$f(x) = \frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(\sin x - \frac{1}{\sin x}\right)}{\sqrt{2}} + c$$

$$f\left(\frac{\pi}{2}\right)=0\Rightarrow c=0$$

$$f\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\frac{1}{\sqrt{2}}-\sqrt{2}}{\sqrt{2}}\right)$$

$$f\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$$

52. If $\int\frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx=A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$, where C is the integration constant, then AB is equal to : **[JEE(Main)Jan.2024]**

यदि $\int\frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx=A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$ है, जहां C समाकलन अचर

g$r ksAB बराबर है : **[JEE(Main)Jan.2024]**

(A) 4 sec θ (B) 4 cosec (2θ) (X) 2 σ χ θ (D) 8 cosec (2θ)

Ans. D

Sol. $I=\int\frac{\left(\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x\right)dx}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}$

$$I=\int\frac{\sec^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}+\int\frac{\text{cosec}^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\sec x\sin(x-\theta)}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\text{cosec}\, x\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\cos\theta-\cot x\sin\theta}}=I_1+I_2$$

$$I_1=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}$$

Let $\tan x\cos\theta-\sin\theta=t^2$

$$\Rightarrow \sec^2 x dx = \frac{2tdt}{\cos\theta}$$

$$I_1 = \int \frac{\frac{2tdt}{\cos\theta}}{t} = \frac{2t}{\cos\theta} + C_1$$

$$= \frac{2}{\cos\theta}\sqrt{\tan x \cot\theta - \sin\theta} + C_1$$

$$I_2 = \frac{2}{\sin\theta}\sqrt{\cos\theta - \sin\theta \cot x} + C_2$$

$$\Rightarrow I_1 + I_2 \Rightarrow B = \frac{2}{\sin\theta}, A = \frac{2}{\cos\theta}$$

$$\Rightarrow AB = \frac{4}{\sin\theta\cos\theta} = 8\,\text{cosec}\,2\theta$$

53. If $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$, where a, b, c are rational numbers, then 2a + 3b – 4c is equal to :

[JEE(Main)Jan.2024]

यदि $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$ है, जहाँ a, b, c परिमेय संख्याएँ हैं, तो 2a + 3b – 4c बराबर है :

[JEE(Main)Jan.2024]

(A) 10 (B) 7 (C) 4 (D) 8

Ans. D

Sol. $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{x} + c\sqrt{3}$

Rationalize

$$\int_0^1 \left(\frac{\sqrt{3+x}-\sqrt{1+x}}{2}\right)dx$$

$$= \frac{1}{2}\left[\frac{2}{3}(3+x)^{\frac{3}{2}} - \frac{2}{3}(1+x)^{3/2}\right]_0^1$$

$$= \frac{1}{3}[(8-2\sqrt{2}) - (3\sqrt{3}-1)]$$

$$= \frac{1}{3}\left[8 - 2\sqrt{2} - 3\sqrt{2} + 1\right]$$

$$= \frac{1}{3}\left[9 - 2\sqrt{2} - 3\sqrt{3}\right]$$

$= 3 - \frac{2}{3}\sqrt{2} - \sqrt{3}$

$= a + b\sqrt{2} + c\sqrt{3}$

$a = 3 \left| b = -\frac{2}{3} \right| c = -1$

$2(3) + 3\left(-\frac{2}{3}\right) - 4(-1)$

$\Rightarrow 6 - 2 + 4$

$\Rightarrow 8$

54. If (a, b) be the orthocentre of the triangle whose vertices are (1,2), (2,3) and (3,1), and $I_1 = \int_a^b x \sin\left(4x - x^2\right) dx$, $I_2 = \int_a^b \sin\left(4x - x^2\right) dx$, then $36\frac{I_1}{I_2}$ is equal to : **[JEE(Main)Jan.2024]**

यदि शीर्षों (1,2), (2,3) तथा (3,1) के त्रिभुज का लंबकेन्द्र (a, b) है, तथा $I_1 = \int_a^b x \sin\left(4x - x^2\right) dx$, $I_2 = \int_a^b \sin\left(4x - x^2\right) dx$ है, तो $36\frac{I_1}{I_2}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 72 (B) 88 (C) 66 (D) 80

Ans. A

Sol. Slope of AB = 1.

Altitude through C

$y - 1 = -1(x - 3)$

$\Rightarrow x + y = 4.$

Orthocentre (a, b) lies on this

$\Rightarrow a + b = 4$ _____(i)

Now, $I_1 = \int_a^b x.\sin\left(4x - x^2\right) dx$

$= \int_a^b (a + b - x) \sin\left(4(a + b - x) - (a + b - x)^2\right) dx$

$= \int_a^b (4 - x) \sin\left(4x - x^2\right) dx$ {using (i)}

$\Rightarrow I_1 = 4 \cdot I_2 - I_1$

$$\Rightarrow \frac{I_1}{I_2} = 2$$

$$\Rightarrow 36\frac{I_1}{I_2} = 72$$

55. For $0 < a < 1$, the value of the integral $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}$ is : **[JEE(Main)Jan.2024]**

$0 < a < 1$ के लिए, समाकलन $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\frac{\pi}{1-a^2}$ (B) $\frac{\pi^2}{\pi+a^2}$ (C) $\frac{\pi^2}{\pi-a^2}$ (D) $\frac{\pi}{1+a^2}$

Ans. A

Sol. $I = \int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}; 0 < a < 1$

$$I = \int_0^{\pi} \frac{dx}{1+2a\cos x + a^2}$$

$$2I = 2\int_0^{\pi/2} \frac{2\left(1+a^2\right)}{\left(1+a^2\right)^2 - 4a^2\cos^2 x}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\sec^2 x - 4a^2}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2\cdot\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\tan^2 x + \left(1-a^2\right)^2}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\frac{2\cdot\sec^2 x}{1+a^2}\cdot dx}{\tan^2 x + \left(\frac{1-a^2}{1+a^2}\right)^2}$$

$$\Rightarrow I = \frac{2}{\left(1-a^2\right)}\left[\frac{\pi}{2} - 0\right]$$

$$I = \frac{\pi}{1-a^2}$$

56. If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is :

[JEE(Main)Jan.2024]

यदि समाकलन $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$ है, तो a का मान है : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{2}$ (B) 3 (C) 2 (D) $-\frac{3}{2}$

Ans. B

Sol. By even odd property

$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\sin^2 x+1\right)dx$

$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\frac{3}{2}-\frac{\cos 2x}{2}\right)dx$

$\left(x^2\sin x-\int 2x\sin x dx+\frac{3x}{2}-\frac{\sin 2x}{4}\right)_0^{\frac{\pi}{2}}$

$=\left[x^2\sin x-\left[2x(-\cos x)-\int 2(-\cos x)dx\right]+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$

$=\left[x^2\sin x+2x\cos x-2\sin x+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$

$=\left[\frac{\pi^2}{4}+0-2+\frac{3\pi}{4}-0\right]-[0]=\frac{\pi^2}{4}+\frac{3\pi}{4}-2$

57. $\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2}\int_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ is equal to : **[JEE(Main)Jan.2024]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{3\pi}{4}$ (B) $\dfrac{3\pi^2}{8}$ (C) $\dfrac{3\pi^2}{4}$ (D) $\dfrac{3\pi}{8}$

Ans. B

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{1}{3}}\right)}{\left(x-\frac{p}{2}\right)^2}=\dfrac{0}{0}$ form

Apply leibniz

$$\lim_{x\to\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}\right)\times 0-\cos(x)\cdot 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$x\to\dfrac{\pi}{2}+t$

$\Rightarrow t\to 0$

$$\lim_{t\to 0}\frac{-\cos\left(\frac{\pi}{2}+t\right)3\left(\frac{\pi}{2}\right)^2}{2t}$$

$$\lim_{t\to 0}\frac{3\pi^2\sin t}{8t}=\frac{3\pi^2}{8}$$

58. The value of $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ is : **[JEE(Main)Jan.2024]**

$\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{\left(2\sqrt{3}+3\right)\pi}{24}$ (B) $\dfrac{\pi}{8\left(2\sqrt{3}+3\right)}$ (C) $\dfrac{13\left(2\sqrt{3}-3\right)\pi}{8}$ (D) $\dfrac{13\pi}{8\left(4\sqrt{3}+3\right)}$

Ans. D

Sol. $$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{n^3}{n^4\left(1+\frac{k^2}{n^2}\right)\left(1+\frac{3k^2}{n^2}\right)}$$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{n^3}{\left(1+\frac{k^2}{n^2}\right)\left(1+\frac{3k^2}{n^2}\right)}$$

$$=\int_0^1\frac{dx}{3\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{3}\times\frac{3}{2}\int_0^1\frac{\left(x^2-1\right)-\left(x^2+\frac{1}{3}\right)}{\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{2}\int_0^1\left[\frac{1}{x^2+\left(\frac{1}{\sqrt{3}}\right)^2}-\frac{1}{1+x^2}\right]dx$$

$$=\frac{1}{2}\left[\sqrt{3}\tan^{-1}\left(\sqrt{3}x\right)\right]_0^1-\frac{1}{2}\left(\tan^{-1}x\right)_0^1$$

$$=\frac{\sqrt{3}}{2}\left(\frac{\pi}{3}\right)-\frac{1}{2}\left(\frac{\pi}{4}\right)$$

$$=\frac{\pi}{2\sqrt{3}}-\frac{\pi}{8}$$

$$=\frac{\pi}{2}\left(\frac{1}{\sqrt{3}}-\frac{1}{4}\right)$$

$$=\frac{\pi}{2}\left(\frac{4-\sqrt{3}}{4\sqrt{3}}\right)\left(\frac{4+\sqrt{3}}{4+\sqrt{3}}\right)$$

$$=\frac{\pi}{2}\left(\frac{16-3}{4\sqrt{3}\left(4+\sqrt{3}\right)}\right)$$

$$= \frac{13\pi}{8\left(4\sqrt{3}+3\right)}$$

59. Let $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim\limits_{x\to 0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$, then $8\alpha^2$ is equal to : **[JEE(Main)Jan.2024]**

माना $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ एक अवकलनीय फलन है तथा $f(0)=\frac{1}{2}$ है। यदि $\lim\limits_{x\to 0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$ है, तो $8\alpha^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 16 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $$\lim_{x\to 0}\frac{\int_0^x f(t)\,dt+xf(x)}{e^{x^2}(2x)}=\lim_{x\to 0}\frac{\int_0^x f(t)\,dt}{e^{x^2}(2x)}+\lim_{x\to 0}\frac{f(x)}{e^{x^2}(2)}$$

$$=\lim_{x\to 0}\frac{f(x)}{\left(e^{x^2}+xe^{x^2}(2x)\right)}+\frac{1}{4}=\frac{1}{2}$$

$$\Rightarrow 8\alpha^2=2$$

60. Let $f: R\to R$ be defined as $f(x)=ae^{2x}+be^x+cx$. If $f(0)=-1, f'(\log_e 2)=21$ and $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$, then the value of $|a+b+c|$ equals : **[JEE(Main)Jan.2024]**

माना Let $f: R\to R$, $f(x)=ae^{2x}+be^x+cx$ द्वारा परिभाषित है। यदि $f(0)=-1, f'(\log_e 2)=21$ तथा $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$ हैं, तो $|a+b+c|$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 12 (B) 16 (C) 8 (D) 10

Ans. C

Sol. $f(x)=ae^{2x}+be^x+cx$ $\qquad f(0)=-1$

$a+b=-1$

$f'(x)=2ae^{2x}+be^x+c$ $\qquad f'(\ln 2)=21$

$8a+2b+c=21$

$$\int_0^{\ell n4}\left(ae^{2x}+be^x\right)dx=\frac{39}{2}$$

$$\left[\frac{ae^{2x}}{2}+be^{x}\right]_{0}^{\ell n4}=\frac{39}{2}\Rightarrow 8a+4b-\frac{a}{2}-b=\frac{39}{2}$$

$15a + 6b = 39$

$15a - 6a - 6 = 39$

$9a = 45 \Rightarrow a = 5$

$b = -6$

$c = 21 - 40 + 12 = -7$

$a + b + c - 8$

$|a + b + c| = 8$

61. Let a and b be real constants such that the function f defined by $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$ be differentiable on R. Then, the value of $\int_{-2}^{2} f(x)dx$ equal : **[JEE(Main)Jan.2024]**

माना वास्तविक अचर a तथा b इस प्रकार हैं कि फलन $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$, R पर अवकलनीय है। तो $\int_{-2}^{2} f(x)dx$ का eku cjkcj gS: **[JEE(Main)Jan.2024]**

(A) 21 (B) $\frac{15}{6}$ (C) 17 (D) $\frac{19}{6}$

Ans. C

Sol. f(x) is continuous at x = 1

$f(1^-) = f(1) = f(1^+)$

$a + 4 = b + 2 \Rightarrow b = a + 2$

f(x) is differential at x = 1

$f'(1^-) = f'(1^+)$

$5 = b \Rightarrow a = 3$

Now $\int_{-2}^{2} f(x)dx = \int_{-2}^{1}\left(x^2+3x+3\right)dx+\int_{1}^{2}(5x+2)dx$

$$=\left(\frac{x^3}{3}+\frac{3x^2}{2}+3x\right)_{-2}^{1}+\left(\frac{5x^2}{2}+2x\right)_{1}^{2}=\left(\frac{1}{3}(1+8)+\frac{3}{2}(1-4)+9\right)+\frac{5}{2}(4-1)+2$$

$$=\frac{1}{3}+\frac{3}{2}+3-\left(\frac{-8}{3}+6-6\right)+\left(10+4-\frac{5}{2}-2\right)$$

$$=6+\frac{3}{2}+12-\frac{5}{2}=18-1=17$$

62. Let y = f(x) be a thrice differentiable function in (–5, 5). Let the tangents to the curve y = f(x) at (1, f(1)) and (3, f(3)) make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive x-axis. If $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ where α, β are integers, then the value of $\alpha+\beta$ equals : **[JEE(Main)Jan.2024]**

माना फलन y = f(x) अंतराल (–5, 5) तीन बार अवकलनीय है। माना वक्र y = f(x) के बिन्दुओं (1, f(1)) तथा (3, f(3)) पर स्पर्श रेखाऐं धनात्मक x-अक्ष से $\frac{\pi}{6}$ तथा $\frac{\pi}{4}$ के कोण बनाती हैं। यदि $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ हैं, जहाँ α, β पूर्णांक हैं, तो $\alpha+\beta$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 26 (B) –16 (C) –14 (D) 36

Ans. A

Sol. $y=f(x)\Rightarrow\frac{dy}{dx}=f'(x)$

$$\left.\frac{dy}{dx}\right)_{(1,f(1))}=f'(1)=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}\Rightarrow f'(1)=\frac{1}{\sqrt{3}}$$

$$\left.\frac{dy}{dx}\right)_{(3,f(3))}=f'(3)=\tan\frac{\pi}{4}=1\Rightarrow f'(3)=1$$

$$27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$$

$$I=\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt$$

$f'(t)=z\Rightarrow f''(t)\,dt=dz$

$z=f'(3)=1$

$$z=f'(1)=\frac{1}{\sqrt{3}}$$

$$I=\int_{\frac{1}{\sqrt{3}}}^{1}\left(z^2+1\right)dz=\left(\frac{z^3}{3}+z\right)_{\frac{1}{\sqrt{3}}}^{1}$$

$$=\left(\frac{1}{3}+1\right)-\left(\frac{1}{3}\cdot\frac{1}{3\sqrt{3}}+\frac{1}{\sqrt{3}}\right)$$

$$=\frac{4}{3}-\frac{10}{9\sqrt{3}}=\frac{4}{3}-\frac{10}{27}\sqrt{3}$$

$$\alpha+\beta\sqrt{3}=27\left(\frac{4}{3}-\frac{10}{27}\sqrt{3}\right)=36-10\sqrt{3}$$

$\alpha = 36, \beta = -10$

$\alpha + \beta = 36 - 10 = 26$

63. Let $f: R \to R$ be a function defined by $f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ and $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$. Then, $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ equal to : **[JEE(Main)Jan.2024]**

माना फलन $f: R \to R$, $f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ द्वारा परिभाषित है तथा $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$ है। तो $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 42 (B) 33 (C) 36 (D) 39

Ans. D

Sol. $f(x) = \frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}$

$$fof(x) = \frac{f(x)}{\left(1+f(x)^4\right)^{\frac{1}{4}}} = \frac{\frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}}{\left(1+\frac{x^4}{1+x^4}\right)^{\frac{1}{4}}} = \frac{x}{\left(1+2x^4\right)^{\frac{1}{4}}}$$

$$f\left(f\left(f\left(f(x)\right)\right)\right) = \frac{x}{\left(1+4x^4\right)^{\frac{1}{4}}}$$

$$18\int_0^{\sqrt{2\sqrt{5}}} \frac{x^3}{\left(1+4x^4\right)^{\frac{1}{4}}}dx$$

Let $1 + 4x^4 = t^4$

$16x^3\,dx = 4t^3\,dt$

$$\frac{18}{4}\int_1^3 \frac{t^3 dt}{t}$$

$$= \frac{9}{2}\left(\frac{t^3}{3}\right)_1^3$$

$$= \frac{3}{2}[26] = 39$$

64. Let a be the sum of all coefficients in the expansion of $(1 - 2x + 2x^2)^{2023}(3 - 4x^2 + 2x^3)^{2024}$ and

$b = \lim\limits_{x\to 0}\left(\dfrac{\int_0^x \frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equtions $cx^2 + dx + e = 0$ and $2bx^2 + ax + 4 = 0$ have a common root, where c, d, e $\in$ R, then d : c : e equals : **[JEE(Main)Jan.2024]**

माना $(1 - 2x + 2x^2)^{2023}(3 - 4x^2 + 2x^3)^{2024}$ के प्रसार में सभी पदों के गुणांक का योग a है तथा $b = \lim\limits_{x\to 0}\left(\dfrac{\int_0^x \frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$ है।

यदि समीकरणों $cx^2 + dx + e = 0$ तथा $2bx^2 + ax + 4 = 0$ का एक उभयनिष्ठ मूल हैं, जहाँ c, d, e $\in$ R हैं, तो d : c : e बराबर gS **[JEE(Main)Jan.2024]**

(A) 1 : 1 : 4 (B) 4 : 1 : 4 (C) 2 : 1 : 4 (D) 1 : 2 : 4

Ans. A

Sol. By putting x = 1

$a = (1 - 2 + 2)^{2023}(3 - 4 + 2)^{2024} = (1)(1) = 1$

$$b = \lim_{x\to 0}\frac{\int_0^x \frac{\ell n(1+t)}{t^{2024}+1}dt}{x^2}\ \left(\frac{0}{0}\text{ form}\right)$$

By DL hospital Rule

$$\lim_{x\to 0}\frac{\frac{\ell n(1+x)}{x^{2024}+1}}{2x}$$

$$= \frac{1}{2}\cdot\frac{1}{(0+1)} = \frac{1}{2}$$

Now $cx^2 + dx + e = 0$ and $x^2 + x + 4 = 0$ have a common root.
But D < 0 for second equation. So both roots are common.

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{1}$$

d : c : e = 1 : 1 : 4

65. Let $f, g : (0,\infty) \to \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^{x}\left(|t| - t^2\right)e^{-t^2}dt$ and $g(x) = \int_0^{x^2} t^{1/2}e^{-t}dt$. Then the value of $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to : **[JEE(Main)Jan.2024]**

माना $f(x)=\int_{-x}^{x}\left(|t|-t^2\right)e^{-t^2}dt$ तथा $g(x)=\int_0^{x^2}t^{1/2}e^{-t}dt$ द्वारा परिभाषित दो फलन $f,g:(0,\infty)\to\mathbb{R}$ हैं। तो $9\left(f\left(\sqrt{\log_e 9}\right)+g\left(\sqrt{\log_e 9}\right)\right)$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 6 (C) 10 (D) 8

Ans. D

Sol. $f(x)=\int_{-x}^{x}\left(|t|-t^2\right)e^{-t^2}dt$

$\Rightarrow f'(x)=2\cdot\left(|x|-x^2\right)e^{-x^2}$ .....(1)

$g(x)=\int_0^{x^2}t^{\frac{1}{2}}e^{-t}dt$

$g'(x)=x^{-x^2}(2x)-0$

$f'(x)+g'(x)=2x^{-x^2}-2x^2e^{-x^2}+2x^2e^{-x^2}$

Integrating both sides w.r.t.x

$f(x)+g(x)=\int_0^{\alpha}2xe^{-x^2}dx$

$x^2=t$

$\Rightarrow\int_0^{\sqrt{\alpha}}e^{-t}dt=\left[-e^{-t}\right]_0^{\sqrt{\alpha}}$

$=-e^{\left(\log_e(9)^{-1}\right)+1}$

$\Rightarrow 9(f(x)+g(x))=\left(1-\frac{1}{9}\right)9=8$

66. The value of the integral $\int_0^{\frac{\pi}{4}}\frac{xdx}{\sin^4(2x)+\cos^4(2x)}$ equals : **[JEE(Main)Jan.2024]**

समाकलन $\int_0^{\frac{\pi}{4}}\frac{xdx}{\sin^4(2x)+\cos^4(2x)}$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{\sqrt{2}\pi^2}{32}$ (B) $\frac{\sqrt{2}\pi^2}{8}$ (C) $\frac{\sqrt{2}\pi^2}{16}$ (D) $\frac{\sqrt{2}\pi^2}{64}$

Ans. A

Sol. By property P–6

$2I=\int_0^{\frac{\pi}{4}}\left(\frac{x}{\cos^4 2x+\sin^4 2x}+\frac{\frac{\pi}{4}-x}{\sin^4 2x+\cos^4 2x}\right)dx \Rightarrow \frac{\pi}{4}\frac{1}{2}\int_0^{\frac{\pi}{4}}\frac{d\theta}{\sin^4\theta+\cos^4\theta}=\frac{\pi}{8}\int_0^1\frac{1+t^2}{t^4+1}dt,$

$t = \tan\theta$

$$= \frac{\pi}{8}\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)_0^1 = \frac{\pi^2}{16\sqrt{2}}$$

67. The value of $\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ is equal to : **[JEE(Main)Jan.2024]**

$\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 0 (D) –1

Ans. C

Sol. $2x^3-3x^2-x+1=\frac{1}{4}(2x-1)\left((2x-1)^2-5\right)$

$$I=\int_0^1\left(\left(\frac{(2x-1)\left((2x-1)^2-5\right)}{4}\right)\right)^{\frac{1}{3}}dx$$

put $2x-1=t$

$$I=\frac{1}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\frac{t\left(t^2-5\right)}{4}\right)^{\frac{1}{3}}dt$$

$= 0$ ($\because$ odd function in t)

68. If $\int_0^{\frac{\pi}{3}}\cos^4x\,dx = a\pi + b\sqrt{3}$, where a and b are rational numbers, then 9a + 8b is equal to : **[JEE(Main)Jan.2024]**

यदि $\int_0^{\frac{\pi}{3}}\cos^4x\,dx = a\pi + b\sqrt{3}$, जहाँ a तथा b परिमेय संख्याएँ है, तो 9a + 8b बराबर होगा : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{2}$ (B) 2 (C) 1 (D) 3

Ans. B

Sol. $\int_0^{\pi/3}\cos^4x dx$

$$=\int_0^{\pi/3}\left(\frac{1+\cos 2x}{2}\right)^2dx$$

$$=\frac{1}{4}\int_0^{\pi/3}\left(1+2\cos 2x+\cos^2 2x\right)dx$$

$$=\frac{1}{4}\left[\int_0^{\pi/3}dx+2\int_0^{\pi/3}\cos 2x dx+\int_0^{\pi/3}\frac{1+\cos 4x}{2}dx\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{3}+(\sin 2x)_0^{\pi/3}+\frac{1}{2}\left(\frac{\pi}{3}\right)+\frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{3}+(\sin 2x)_0^{\pi/3}+\frac{1}{2}\left(\frac{\pi}{3}\right)+\frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{2}+\frac{\sqrt{3}}{2}+\frac{1}{8}\times\left(-\frac{\sqrt{3}}{2}\right)\right]$$

$$=\frac{\pi}{2}+\frac{7\sqrt{3}}{64}$$

$$\therefore a=\frac{1}{8};b=\frac{7}{64}$$

$$\therefore 9a+8b=\frac{9}{8}+\frac{7}{8}=2$$

69. Let $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function.

If $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to _____. **[JEE(Main)Jan.2024]**

माना $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$ है, जहाँ g एक संतत विषम फलन है। यदि $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$ है,

तो α बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. B

Sol. $f(x)=\int_0^x g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$

$$f(-x)=\int_0^{-x} g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$$

$$f(-x)=-\int_0^x g(-y)\ell n\left(\frac{1+y}{1-y}\right)dy$$

$$=-\int_0^x g(y)\ell n\left(\frac{1-y}{1+y}\right)dy \text{ (g is odd)}$$

$f(-x) = -f(x) \Rightarrow f$ is also odd

Now,

$$I = \int_{-\pi/2}^{\pi/2} \left(f(x) + \frac{x^2 \cos x}{1+e^x} \right) dx$$

$$I = \int_{-\pi/2}^{\pi/2} \left(f(-x) + \frac{x^2 e^x \cos x}{1+e^x} \right) dx$$

$$2I = \int_{-\pi/2}^{\pi/2} x^2 \cos x dx = 2\int_0^{\pi/2} x^2 \cos x dx$$

$$I = \left(x^2 \sin x \right)_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x dx$$

$$= \frac{\pi^2}{4} - 2\left(-x\cos x + \int \cos x dx \right)_0^{\pi/2}$$

$$= \frac{\pi^2}{4} - 2(0+1) = \frac{\pi^2}{4} - 2 \Rightarrow \left(\frac{\pi}{2} \right)^2 - 2$$

$\therefore$ a = 2

70. Let the slope of the line 45x + 5y + 3 = 0 be $27r_1 + \frac{9r_2}{2}$ for some $r_1, r_2 \in R$. Then

$$\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$$ is equal to _____. **[JEE(Main)Jan.2024]**

माना किसी $r_1, r_2 \in R$ के लिए रेखा 45x + 5y + 3 = 0 की प्रवणता $27r_1 + \frac{9r_2}{2}$ है। तो $\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$

बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 12

Sol. Slope of 45x + 5y + 3 = 0 is m = – 9

$$\therefore 27r_1 + \frac{9r_2}{2} = -9$$

$$\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$$

$$\lim_{x\to 3}\frac{\int_3^x 8t^2dt}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}$$

$$= \lim_{x\to 3}\frac{8x^2}{\frac{3r_2}{2}-2r_2x-3r_1x^2-3} \text{ by L.H. rule}$$

$$=\frac{8\times 9}{\frac{3r_2}{2}-6r_2-27r_1-3}$$

$$=\frac{8\times 9}{-\frac{9r_2}{2}-27r_1-3}$$

$$=\frac{8\times 9}{6}=12$$

71. If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, जहाँ α, β तथा γ परिमेय संख्याऐं हैं, तो $3\alpha + 4\beta - \gamma$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 6

Sol. $=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}\,dx$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}|\sin x-\cos x|\,dx$$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}(\cos x-\sin x)dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}(\sin x-\cos x)dx$$

$$=-1+2\sqrt{2}-\sqrt{3}$$

$$=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$$

$\alpha = -1$, $\beta = 2$, $\gamma = -1$

$3\alpha + 4\beta - \gamma = 6$

72. The value of $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$, where [t] denotes the greatest integer less than or equal to t, is _____.

[JEE(Main)Jan.2024]

$9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$, जहाँ [t] महत्तम पूर्णांक $\le t$ है, का मान है _____। **[JEE(Main)Jan.2024]**

Ans. 155

Sol. $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$

Let $I=\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$

$\frac{10x}{x+1}=1\Rightarrow x=\frac{1}{9}$

$\frac{10x}{x+1}=4 \qquad \Rightarrow x=\frac{2}{3}$

$\frac{10x}{x+1}=9 \qquad \Rightarrow x=9$

$I=\int_0^{\frac{1}{9}}0dx+\int_{\frac{1}{9}}^{\frac{2}{3}}1dx+\int_{\frac{2}{3}}^{9}2dx$

$=0+\left[x\right]_{\frac{1}{9}}^{\frac{2}{3}}+\left[2x\right]_{\frac{2}{3}}^{9}$

$I=\frac{2}{3}-\frac{1}{9}+18-\frac{4}{3}$

$\Rightarrow I=\frac{155}{9}$

$\Rightarrow 9I=155$

73. Let $f:R\to R$ be a function defined by $f(x)=\frac{4^x}{4^x+2}$ and $M=\int_{f(a)}^{f(1-a)}x\sin^4\left(x(1-x)\right)dx$,

$N=\int_{f(a)}^{f(1-a)}\sin^4\left(x(1-x)\right)dx$; $a\neq\frac{1}{2}$. If $\alpha M=\beta N$, $\alpha,\beta\in N$, then the least value of $\alpha^2+\beta^2$ is equal to _____.

[JEE(Main)Jan.2024]

माना फलन $f: R \to R$, $f(x)=\frac{4^x}{4^x+2}$ द्वारा परिभाषित है तथा $M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$,

$N=\int\limits_{f(a)}^{f(1-a)} \sin^4\left(x(1-x)\right)dx$; $a \neq \frac{1}{2}$ हैं। यदि $\alpha M = \beta N$, $\alpha, \beta \in N$ हैं, तो $\alpha^2+\beta^2$ का न्यूनतम मान है _____.

[JEE(Main)Jan.2024]

Ans. 5

Sol. $f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{4^{1-x}}{4^{1-x}}$

$=\frac{4^x}{4^x+2}+\frac{4}{4+2\cdot 4^x}$

$=\frac{4^x+2}{4^x+2}=1$

$\Rightarrow M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$

$x \to f(a)+f(1-a)-x=1-x$

$M=\int\limits_{f(a)}^{f(1-a)} (1-x)\sin^4\left((1-x)x\right)dx$

$M=N-M \Rightarrow \frac{M}{N}=\frac{1}{2}=\frac{\beta}{\alpha}$

$\Rightarrow \left(\alpha^2+\beta^2\right)_{least}=1+4=5$

74. If the integral $525\int\limits_0^{\frac{\pi}{2}} \sin 2x\cos^{\frac{11}{2}}x\left(1+\cos^{\frac{5}{2}}x\right)^{\frac{1}{2}}dx$ is equal to $\left(n\sqrt{2}-64\right)$, then n is equal to _____.

[JEE(Main)Jan.2024]

यदि समाकलन $525\int\limits_0^{\frac{\pi}{2}} \sin 2x\cos^{\frac{11}{2}}x\left(1+\cos^{\frac{5}{2}}x\right)^{\frac{1}{2}}dx=\left(n\sqrt{2}-64\right)$ है, तो n बराबर है_____.

[JEE(Main)Jan.2024]

Ans. 176

Sol. $I = 525\int \sin 2x \cos^{\frac{11}{2}} x\left(1+\cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$

put $\sqrt{\cos x} = t \Rightarrow \cos x = t^2$

$-\sin x\, dx = 2t dt$

$\sin x\, dx = -2t\, dt$

$$I = 525\int_0^1 2(2t)t^2 t^{11}\sqrt{1+t^5}\,dt$$

$$= 525\times 4\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

$$= 2100\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

again put $\sqrt{1+t^5} = u \quad \Rightarrow 1+t^5 = u^2$

$5t^4dt = 2udu$

$$I = 2100\int_1^{\sqrt{2}} u\cdot\frac{2u}{5}\cdot\left(u^2-1\right)^2 du$$

$$= 420\times 2\left(\int_1^{\sqrt{2}} u^2\left(u^4-2u^2+1\right)du\right)$$

$$= 840\int_1^{\sqrt{2}}\left(u^6-2u^4+u^2\right)du$$

$$= 840\left[\frac{u^7}{7}-\frac{2u^5}{5}+\frac{u^3}{3}\right]_1^{\sqrt{2}}$$

$$= 840\left[\frac{8\sqrt{2}}{7}-\frac{8\sqrt{2}}{5}+\frac{2\sqrt{2}}{3}-\frac{1}{7}+\frac{2}{5}-\frac{1}{3}\right]$$

$= 176\sqrt{2}-64$

Compare with $n\sqrt{2}-64$

$n = 176$

75. $\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx\right|$ is equal to _____. **[JEE(Main)Jan.2024]**

$\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx\right|$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 15

Sol. $I=\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$2I=\int_0^{\pi}\frac{\sin x\cos x}{\sin^4 x+\cos^4 x}\left(x^2-(\pi-x)^2\right)dx$

$2I=\int_0^{\pi}\frac{\pi(2x-\pi)\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$\Rightarrow \quad I=\pi\int_0^{\pi/2}\frac{(2x-\pi)\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$\Rightarrow \quad I=\pi\int_0^{\pi/2}\frac{-2x\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$2I=-\pi^2\int_0^{\pi/2}\frac{\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$I=\frac{-\pi}{2\times 2}\int_0^{\pi/2}\frac{\sin 2x}{\sin^4 x+\cos^4 x}dx$

$I=\frac{-\pi^2}{4}\int_0^{\pi/2}\frac{2\tan x\sec^2 x}{\tan^4 x+1}dx$

$I=\frac{-\pi^2}{4}\tan^{-1}\left(\tan^2 x\right)_0^{\pi/2}$

$\Rightarrow \quad I=\frac{-\pi^3}{8}$

$\therefore \quad \left|\frac{120}{\pi^3}I\right|=15$

76. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}=\alpha\pi+\beta\log_e\left(3+2\sqrt{2}\right)$, where α, β are integers, then $\alpha^2+\beta^2$ equals _____.

[JEE(Main)Jan.2024]

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}=\alpha\pi+\beta\log_e\left(3+2\sqrt{2}\right)$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha^2+\beta^2$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 8

Sol. $$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}dx \qquad ...(1)$$

Apply king

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x\left(e^{\sin x}\right)}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}dx \qquad ...(2)$$

adding (1) & (2)

$$2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$$I=\int_{0}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$\sin x = t$

$$I=\int_{0}^{1}\frac{8\sqrt{2}}{1+t^4}dx$$

$$I=4\sqrt{2}\int_{0}^{1}\left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}-\frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}\right)dt$$

$$I=4\sqrt{2}\int_{0}^{1}\frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}-\frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}\right)^2-2}dt$$

Let $t-\frac{1}{t}=z$ & $t+\frac{1}{t}=k$

$$=4\sqrt{2}\left[\int_{-\infty}^{0}\frac{dz}{z^2+2}-\int_{\infty}^{2}\frac{dk}{k^2-2}\right]$$

$$=4\sqrt{2}\left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{z}{\sqrt{2}}\right]_{-\infty}^{0}-\left[\frac{1}{2\sqrt{2}}\ell n\left(\frac{k-\sqrt{2}}{k+\sqrt{2}}\right)\right]_{\infty}^{2}$$

$$= 4\sqrt{2}\left[\frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\left[\ell n\frac{2-\sqrt{2}}{2+\sqrt{2}}\right]\right]$$

$$= 2\pi + 2\ell n\left(3+2\sqrt{2}\right)$$

$\alpha = 2$

$\beta = 2$

77. Let $f: (0, \infty) \to R$ and $F(x) = \int_0^x t\,f(t)\,dt$. If $F(x^2) = x^4 + x^5$, then $\sum_{r=1}^{12} f\left(r^2\right)$ is equal to _____.

[JEE(Main)Jan.2024]

माना $f: (0, \infty) \to R$ तथा $F(x) = \int_0^x t\,f(t)\,dt$ हैं। यदि $F(x^2) = x^4 + x^5$ है, तो $\sum_{r=1}^{12} f\left(r^2\right)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 219

Sol. $F(x) = \int_0^x tf(t)dt$ and $F(x)^2 = x^4 + x^5$

$F'(x) = xf(x)$ and $2xF'\left(x^2\right) = 4x^3 + 5x^4$

$2F'\left(x^2\right) = 4x^2 + 5x^3$

$2F'(x) = 4x + 5x^{\frac{3}{2}}$

$\Rightarrow 2x + \frac{5}{2}x^{\frac{3}{2}} = xf(x)$

$\Rightarrow f(x) = 2 + \frac{5}{2}x^{\frac{1}{2}}$

$\Rightarrow f\left(x^2\right) = 2 + \frac{5}{2}x$

$$\sum_{r=1}^{12} f\left(r^2\right) = \sum_{r=1}^{12}\left(2+\frac{5}{2}r\right)$$

$$= 24 + \frac{5}{2}\sum_{r=1}^{12} r$$

$$= 24 + \frac{5}{2}\times\frac{12\times13}{2}$$

$24 + 195 = 219$

78. If $\int \frac{1}{a^2\sin^2 x + b^2\cos^2 x}dx = \frac{1}{12}\tan^{-1}(3\tan x) +$ constant, then the maximum value of $a\sin x + b\cos x$, is :

यदि $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) +$ अचर है, तो $\sin x + b \cos x$ का उच्चतम मान है :

[JEE(Main)April,2024]

(A) $\sqrt{40}$ (B) $\sqrt{42}$ (C) $\sqrt{39}$ (D) $\sqrt{41}$

Ans. A

Sol. $\int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$

let $\tan x = t$

$\sec^2 dx = dt$

$\int \frac{dt}{a^2 t^2 + b^2}$

$\frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a}\right)^2}$

$\frac{1}{a^2} \frac{1}{\frac{b}{a}} \tan^{-1}\left(\frac{t}{b} a\right) + c$

$\frac{1}{ab} \tan^{-1}\left(\frac{\alpha}{b} \tan x\right) + c$

on compairing $\frac{a}{b} = 3$

$ab = 12$

$a = 6, b = 2$

maximum value of

$6 \sin x + 2 \cos x$ is $\sqrt{40}$

79. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to : [JEE(Main)April,2024]

माना $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$ है। यदि $I(0) = 3$, तब $I\left(\frac{\pi}{12}\right)$ बराबर होगा : [JEE(Main)April,2024]

(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $3\sqrt{3}$ (D) $6\sqrt{3}$

Ans. C

Sol. $I(x)=\int \frac{6dx}{\sin^2(1-\cot x)^2}$ $\qquad I(0)=3$

$I\left(\frac{\pi}{12}\right)=?$

$$I(x)=\int \frac{6dx}{\sin^2 x\frac{(\sin x-\cos x)^2}{\sin^2 x}}=\int \frac{6dx}{\sin^2 x+\cos^2 x-2\sin\cos x}$$

$$=\int \frac{6dx}{1-2\sin x\cos x}=\int \frac{6\sec^2 xdx}{1+\tan^2 x-2\tan x}$$

$$=\int \frac{6\sec^2 xdx}{(\tan x-1)^2}$$

$= \tan x = t$

$= \sec^2 x\, dx = dt$

$$=\int \frac{6dt}{(t-1)^2}=6\int (t-1)^{-2}\,dt$$

$$I(x)=\frac{-6}{(t-1)}+C$$

$$I(x)=-\frac{6}{\tan x-1}+C$$

$I(0) = 3$

$3 = 6 + C$

$C = -3$

$$I(x)=-\frac{6}{\tan x-1}-3$$

$$I\left(\frac{\pi}{12}\right)=-\frac{6}{2-\sqrt{3}-1}-3$$

$$=-\frac{6}{1-\sqrt{3}}-3=-\left(\frac{6+3-3\sqrt{3}}{1-\sqrt{3}}\right)$$

$$=-\left(\frac{9-3\sqrt{3}}{1-\sqrt{3}}\times\frac{1+\sqrt{3}}{1+\sqrt{3}}\right)$$

$$=-\left(\frac{9+9\sqrt{3}-3\sqrt{3}-9}{1-3}\right)=\frac{6\sqrt{3}}{2}=3\sqrt{3}$$

80. Let $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}\left(\alpha x + \log_e|\beta \sin x + \gamma \cos x|\right) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to : **[JEE(Main)April,2024]**

माना $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}\left(\alpha x + \log_e|\beta \sin x + \gamma \cos x|\right) + C$ है, जहाँ C समाकलन अचर है। तो $\alpha + \frac{\gamma}{\beta}$ बराबर है : **[JEE(Main)April,2024]**

(A) 3 (B) 1 (C) 7 (D) 7

Ans. D

Sol. $\int \frac{2-\tan x}{3+\tan x}dx = \int \frac{2\cos x - \sin x}{3\cos x + \sin x}dx$

Put, $2\cos x - \sin x = a(-3\sin x + \cos x) + b(3\cos x + \sin x)$

$a + 3b = 2$ .....(i)

$-3a + b = -1$ .....(ii)

From equation (i) and (ii) $a = b = \frac{1}{2}$

$$I = \frac{1}{2}\int \frac{-3\sin x + \cos x}{3\cos x + \sin x}dx + \frac{1}{2}\int \frac{3\cos x + \sin x}{3\cos x + \sin x}dx$$

$$I = \frac{1}{2}\ell n|3\cos x + \sin x| + \frac{1}{2}x = \frac{1}{2}\left(\alpha x + \log|\beta \sin x + \gamma \cos x|\right)$$

On comparing, we get

$\alpha = 1, \beta = 1, \gamma = 3$

$$\alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

81. If $\int \text{cosec}^5 x dx = \alpha \cot x \,\text{cosec}\, x\left(\text{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e\left|\tan\frac{x}{2}\right| + C$ where $\alpha, \beta \in R$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____. **[JEE(Main)April,2024]**

यदि $\int \text{cosec}^5 x dx = \alpha \cot x \,\text{cosec}\, x\left(\text{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e\left|\tan\frac{x}{2}\right| + C$ है, जहाँ $\alpha, \beta \in R$ तथा C समाकल अचर है, तो $8(\alpha + \beta)$ का मान बराबर है _____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $\int \text{cosec}^3 x \cdot \text{cosec}^2 x dx = I$

By applying integration by parts

$$I = -\cot x\,\text{cosec}^3 x + \int \cot x\left(-3\text{cosec}^2 x \cot x \,\text{cosec}\, x\right)dx$$

$I = -\cot x \operatorname{cosec}^3 x - 3I - 3\int \operatorname{cosec}^3 x\left(\operatorname{cosec}^2 x - 1\right)dx$

$I = -\cot x \operatorname{cosec}^3 x - 3I + 3\int \operatorname{cosec}^3 x dx$

Let

$I_1 = \int \operatorname{cosec}^3 x dx = -\operatorname{cosec} x \cot x - \int \cot^2 x \operatorname{cosec} x dx$

$I_1 = -\operatorname{cosec} x \cot x - \int\left(\operatorname{cosec}^2 x - 1\right)\operatorname{cosec} x dx$

$2I_1 = -\operatorname{cosec} x \cot x + \ell n\left|\tan\frac{x}{2}\right|$

$I_1 = -\frac{1}{2}\operatorname{cosec} x \cot x + \frac{1}{2}\ell n\left|\tan\frac{x}{2}\right|$

$4I = -\cot x \operatorname{cosec}^3 x - \frac{3}{2}\operatorname{cosec} x \cot x + \frac{3}{2}\ell n\left|\tan\frac{x}{2}\right| + 4c$

$I = -\frac{1}{4}\operatorname{cosec} x \cot x\left(\operatorname{cosec}^2 x + \frac{3}{2}\right) + \frac{3}{2}\ell n\left|\tan\frac{x}{2}\right| + c$

$\therefore \alpha = \frac{-1}{4},\ \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$

82. If $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$, where C is the contant of integration, then the value of $\alpha + \beta +$ 20AB is _____. **[JEE(Main)April,2024]**

यदि $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$ है, जहाँ C समाकलन अचर है, तो $\alpha + \beta + 20AB$ का मान है_____.

[JEE(Main)April,2024]

Ans. 7

Sol. $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$

$I = \int \frac{1}{(x-1)^{\frac{4}{5}}(x+3)^{\frac{6}{5}}}dx$

$$I=\int \frac{1}{\left(\frac{x-1}{x+3}\right)^{\frac{4}{5}}(x+3)^2}dx$$

$\left(\frac{x-1}{x+3}\right)=t$ $\qquad$ $\frac{4}{(x+3)^2}dx=dt$

$$I=\frac{1}{4}\int \frac{1}{t^{\frac{4}{5}}}dt=\frac{1}{4}\left(\frac{t^{\frac{1}{5}}}{\frac{1}{5}}\right)+c$$

$$I=\frac{5}{4}\left(\frac{x-1}{x+3}\right)^{\frac{1}{5}}+C$$

$A=\frac{5}{4}$ $\qquad$ $\alpha=\beta=1$ $\qquad$ $B=\frac{1}{5}$

$$\alpha+\beta+20AB=2+20\times\frac{5}{4}\times\frac{1}{5}=7$$

83. Let $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ and $h(x)=f(|x|)+|f(x)|$. Then $\int_{-2}^{2} h(x)dx$ is equal to :

[JEE(Main)April,2024]

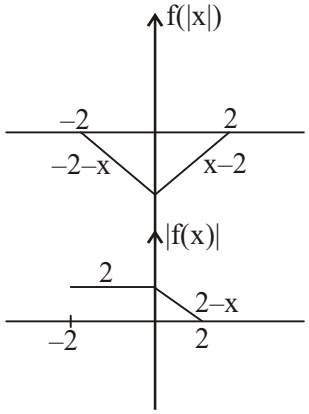
माना $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ तथा $h(x)=f(|x|)+|f(x)|$ हैं। तो $\int_{-2}^{2} h(x)dx$ बराबर है :

[JEE(Main)April,2024]

(A) 2 $\qquad$ (B) 6 $\qquad$ (C) 4 $\qquad$ (D) 1

Ans. A

Sol.



$$\therefore h(x)=\begin{cases}-x & x\in[-2,0]\\ 0 & x\in[0,2]\end{cases}$$

$$\int_{-2}^{2} h(x)dx=\int_{-2}^{0}(-x)dx$$

$$=\left(-\frac{x^2}{22}\right)_{-2}^{0}$$

$$=-(-2)=2$$

84. If the value of the integral $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ is $\frac{2}{\pi}$. Then, a value of α is : **[JEE(Main)April,2024]**

यदि समाकलन $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ का मान $\frac{2}{\pi}$ है। तो α का एक मान है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. Let $I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^x}dx$ _____(1)

$$I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^{-x}}dx$$

(using $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$) _____(2)

Add (1) and (2)

$$2I = \int_{-1}^{+1} \cos(\alpha x)dx = 2\int_{0}^{1} \cos(\alpha x)dx$$

$$I = \frac{\sin\alpha}{\alpha} = \frac{2}{\pi} \text{ (given)}$$

$$\therefore \alpha = \frac{\pi}{2}$$

85. Let $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$. Then, $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ is equal to : **[JEE(Main)April,2024]**

माना $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$ है। तो $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

Ans. B

Sol. $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$

Using L Hopital Rule.

$$\lim_{x\to 0} \frac{f'(x)}{3x^2} = \lim_{x\to 0} \frac{x + \sin\left(1 - e^x\right)}{3x^2} \text{ (Again L hospital)}$$

Using L Hopital Rule

$$\lim_{x\to 0} \frac{-\left[\sin\left(1 - e^x\right)\left(-e^x\right)\cdot e^x + \cos\left(1 - e^x\right)\cdot e^x\right]}{6}$$

$$= -\frac{1}{6}$$

86. The integral $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ is equal to : **[JEE(Main)April,2024]**

समाकलन $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ बराबर है : **[JEE(Main)April,2024]**

(A) $3\pi - 30\log_e 2 + 20\log_e 5$ (B) $3\pi - 50\log_e 2 + 20\log_e 5$

(C) $3\pi - 10\log_e\left(2\sqrt{2}\right) + 10\log_e 5$ (D) $3\pi - 25\log_e 2 + 10\log_e 5$

Ans. B

Sol. $(136)\int_0^{\pi/4}\frac{\sin x}{3\sin x+5\cos x}dx$

$136\sin x=\lambda(3\sin x+5\cos x)+\mu(3\cos x-5\sin x)$

Compare coffi of sinx, cosx

$136=3\lambda-5\mu$ ...(1)

$0=5\lambda+3\mu$ ...(2)

$408=34\lambda\Rightarrow\lambda=12$

$\mu=-20$

$(136)\int_0^{\pi/4}\frac{\sin x}{3\sin x+5\cos x}dx$

$=12(x)_0^{\pi/4}-20\left[\ell n(3\sin x+5\cos x)\right]_0^{\pi/4}$

$=12\frac{\pi}{4}-20\left[\ell n4\sqrt{2}-\ell n5\right]$

$3\pi-20\ell n\left(4\sqrt{2}\right)+20\ell n5$

$=3\pi-50\ell n2+20\ell n5$

87. The value of $\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$ is : **[JEE(Main)April,2024]**

$\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) π^2 (B) $\frac{\pi}{2}$ (C) $\frac{\pi^2}{2}$ (D) $2\pi^2$

Ans. A

Sol. $=\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$

$=\int_0^{\pi}\left(\frac{2y(1+\sin y)}{1+\cos^2 y}+\frac{-2y(1-\sin y)}{1+\cos^2 y}\right)dy$

$=\int_0^{\pi}\left(\frac{2y+2y\sin y-2y+2y\sin y}{1+\cos^2 y}\right)dy$

$I=4\int_0^{\pi}\left(\frac{y\sin y}{1+\cos^2 y}\right)dy$ .....(1)

$$I = 4\int_0^{\pi}\left(\frac{(\pi-y)\sin(\pi-y)}{1+\cos^2(\pi-y)}\right)dy$$

$$I = 4\left[\int_0^{\pi}\left(\frac{\pi\sin y}{1+\cos^2 y}\right)dy - \int_0^{\pi}\frac{y\sin y}{1+\cos^2 y}dy\right] \quad .....(2)$$

Adding equation (1) and (2)

$$2I = 4\int_0^{\pi}\left(\frac{\pi\sin y}{1+\cos^2 y}\right)dy$$

$$I = 2\pi\int_0^{\pi}\left(\frac{\sin y}{1+\cos^2 y}\right)dy$$

$$= 2\pi\times\frac{\pi}{2}$$

$$= \pi^2$$

88. Let $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$, m, n > 0 . If $\int_0^1 (1-x^{10})^{20}dx = a\times\beta(b,c)$, then 100 (a + b + c) equals :

[JEE(Main)April,2024]

माना $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$, m, n > 0 है। यदि $\int_0^1 (1-x^{10})^{20}dx = a\times\beta(b,c)$ है, तो 100 (a + b + c) बराबर है :

[JEE(Main)April,2024]

(A) 1021 (B) 2120 (C) 2012 (D) 1120

Ans. B

Sol. $I = \int_0^1 (1-x^{10})^{20}\,dx$

$x^{10} = t \Rightarrow x = (t)^{1/10}$

$$I = \frac{1}{10}\int_0^1 (t)^{-9/10}(1-t)^{20}\,dt$$

$$a = \frac{1}{10},\ b = \frac{1}{10},\ c = 21$$

$$100\left(\frac{1}{10}+\frac{1}{10}+21\right)$$

$\Rightarrow$ 10 + 10 + 2100 = 2120

89. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$ is equal to : **[JEE(Main)April,2024]**

$\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$ बराबर है : **[JEE(Main)April,2024]**

(A) 1/9 (B) 1/6 (C) 1/3 (D) 1/12

Ans. B

Sol. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$

$\Rightarrow \int_0^{\pi/4} \frac{\tan^2 x \cdot \sec^2 x}{\left(1+\tan^3 x\right)^2} \cdot dx$

Let $1 + \tan^3 x = t$

$3 \cdot \tan^2 x \cdot \sec^2 x \cdot dx = dt$

$\Rightarrow \frac{1}{3}\int_1^2 \frac{dt}{\left(t^2\right)^2}$

$\Rightarrow \frac{1}{3}\left[-\frac{1}{t}\right]_1^2$

$\Rightarrow -\frac{1}{3}\left[+\frac{1}{2}-1\right] \Rightarrow \frac{1}{6}$

90. The value of $k \in N$ for which the integral $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$, satisfies $147\, I_{20} = 148\, I_{21}$ is :

[JEE(Main)April,2024]

$k \in N$ का मान जिसके लिए समाकलन $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$ एवं $147\, I_{20} = 148\, I_{21}$ को संतुष्ट करता है :

[JEE(Main)April,2024]

(A) 8 (B) 7 (C) 10 (D) 14

Ans. B

Sol. $I_n = \int_0^1 \left(1-x^k\right)^n dx$

$$I_{21} = \int_0^1 \left(1 - x^k\right)^{21} dx$$

$$\Rightarrow \int_0^1 \left(1 - x^k\right)\left(1 - x^k\right)^{20} dx$$

$$\Rightarrow \int_0^1 \left(1 - x^k\right)^{20} dx - \int_0^1 x^k \left(1 - x^k\right)^{20} dx$$

$$I_{21} = I_{20} - \int_0^1 x^k \left(1 - x^k\right)^{20} dx$$

$$\int_0^1 xx^{(k-1)} \left(1 - x^k\right)^{20} dx$$

$1 - x^k = t$

$-K(x^{K-1}\, dx) = dt$

Integration by part

$$x\left[-\frac{\left(1 - x^K\right)^{21}}{21K}\right]_0^1 - \frac{1}{21K}\int_0^1 \left(1 - x^k\right)^{21} dx$$

$$I_{21} = I_{20} + 0 - \frac{I_{21}}{21K}$$

$$= \left(\frac{21K + 1}{21K}\right) I_{21} = I_{20}$$

$= (21\ K + 1)\ I_{21} = 21KI_{20}$

$K = 7$

91. Let $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation : **[JEE(Main)April,2024]**

माना $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$ है। तो e^{α} तथा $e^{-\alpha}$ निम्न में से किस समीकरण के मूल हैं : **[JEE(Main)April,2024]**

(A) $2x^2 - 5x - 2 = 0$ (B) $x^2 - 2x - 8 = 0$ (C) $2x^2 - 5x + 2 = 0$ (D) $x^2 + 2x - 8 = 0$

Ans. C

Sol. $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$

Let $e^x - 1 = t^2$

$e^x\, dx = 2t\,dt$

$$\int \frac{2t\,dt}{e^x \cdot t} = \int \frac{2dt}{t^2 + 1}$$

$= 2\tan^{-1} t$

$$= 2\tan^{-1}\left(\sqrt{e^x - 1}\right)_{\alpha}^{\log_e 4}$$

$$= 2\left[\tan^{-1}\sqrt{3} - \tan^{-1}\sqrt{e^{\alpha} - 1}\right] = \frac{\pi}{6}$$

$$= \frac{\pi}{3} - \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{12}$$

$$= \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{4}$$

$e^{\alpha} = 2, \qquad e^{-\alpha} = \frac{1}{2}$

$$x^2 - \left(2 + \frac{1}{2}\right)x + 1 = 0$$

$2x^2 - 5x + 2 = 0$

92. $\lim\limits_{x \to \frac{\pi}{2}} \left(\dfrac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin\left(2t^{\frac{1}{3}}\right) + \cos\left(t^{\frac{1}{3}}\right) \right) dt}{\left(x - \frac{\pi}{2}\right)^2} \right)$ is equal to : **[JEE(Main)April,2024]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\left(\sin\left(2t^{\frac{1}{3}}\right)+\cos\left(t^{\frac{1}{3}}\right)\right)dt}{\left(x-\frac{\pi}{2}\right)^2}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\dfrac{11\pi^2}{10}$ (B) $\dfrac{3\pi^2}{2}$ (C) $\dfrac{5\pi^2}{9}$ (D) $\dfrac{9\pi^2}{8}$

Ans. D

Sol. L. hopital's rule

$$\Rightarrow \lim_{x\to\frac{\pi}{2}}\frac{-3x^2\cdot(\sin 2x+\cos x)}{2\left(x-\frac{\pi}{2}\right)}=\lim_{x\to\frac{\pi}{2}}\frac{-3x^2\cos x(2\sin x+1)}{2\left(x-\frac{\pi}{2}\right)}$$

$$=\lim_{x\to\frac{\pi}{2}}\frac{-9\times\frac{\pi^2}{4}}{2}\times\frac{\cos x}{x-\frac{\pi}{2}}=\lim_{x\to\frac{\pi}{2}}\frac{-9\pi^2}{8}\times\frac{-\sin x}{1}$$

$$=\frac{9\pi^2}{8}$$

93. The integral $\int\limits_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal to : **[JEE(Main)April,2024]**

समाकलन $\int\limits_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ बराबर है : **[JEE(Main)April,2024]**

(A) $\dfrac{1}{4}$ (B) $-\dfrac{1}{2}$ (C) $-\dfrac{1}{4}$ (D) $\dfrac{1}{2}$

Ans. C

Sol. $\int_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$

$x = \cos 2\theta$

$\therefore \quad dx = (-2 \sin 2\theta \, d\theta)$

Tokε λμ ιτ οασα and β

$$-2\int_{\alpha}^{\beta} \cos 2\theta \cdot \sin 2\theta d\theta$$

$$= \int_{\alpha}^{\beta} \sin 4\theta d\theta$$

$$= \frac{-\cos 4\theta}{4}\Bigg|_{\alpha}^{\beta}$$

$$= -\frac{1}{4}\left(2\cdot\left(x^2\right)-1\right)\Bigg|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(2x^2-1\right)\Bigg|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(\frac{18}{16}-1-\frac{2}{16}+1\right)$$

$$= -\frac{1}{4}$$

94. The value of the integral $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$ is : **[JEE(Main)April,2024]**

समाकलन $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$ का मान है : **[JEE(Main)April,2024]**

(A) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$ (B) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

(C) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$ (D) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

Ans. C

Sol. $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$

$$= \left[x\log_e\left(x+\sqrt{x^2+1}\right)\right]_{-1}^{2} - \int_{-1}^{2} \frac{x}{\left(x+\sqrt{x^2+1}\right)}\left(1+\frac{x}{\sqrt{x^2+1}}\right)dx$$

$$= 2\log_e(2+\sqrt{5}) + \log_e(\sqrt{2}-1) - \int_{-1}^{2} \frac{x}{\sqrt{x^2+1}}dx$$

$$= \log_e\left[(2+\sqrt{5})^2(\sqrt{2}-1)\right] - \left[\sqrt{x^2+1}\right]_{-1}^{2}$$

$$= \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right] - \sqrt{5} + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{5} + \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right]$$

95. If the shortest distance between the lines $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ and $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$, and $\int_0^k \left[x^2\right]dx = \alpha - \sqrt{\alpha}$, where [x] denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

[JEE(Main)April,2024]

यदि रेखाओं $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ तथा $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ के बीच न्यूनतम $\frac{38}{3\sqrt{5}}k$ है तथा $\int_0^k \left[x^2\right]dx = \alpha - \sqrt{\alpha}$ है, जहाँ [x] महत्तम पूर्णांक फलन है, तो $6\alpha^3$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 48

Sol. $L_1 \equiv \frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$

$L_2 \equiv \frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$

Shortest distance between L_1 and L_2 is $= \frac{\begin{vmatrix} 5 & 5 & -9 \\ 2 & 3 & 4 \\ 1 & -3 & 2 \end{vmatrix}}{\sqrt{(18)^2 + (0)^2 + (-9)^2}}$

$$\frac{38}{3\sqrt{5}}k = \frac{90+0+81}{9\sqrt{5}} = \frac{19}{\sqrt{5}}$$

$$\Rightarrow k = \frac{3}{2}$$

Now $\int_0^k \left[x^2\right]dx = \int_0^{\frac{3}{2}} \left[x^2\right]dx$

$$= \int_0^1 \left[x^2\right] dx + \int_1^{\sqrt{2}} \left[x^2\right] dx + \int_{\sqrt{2}}^{\frac{3}{2}} \left[x^2\right] dx$$

$$= 0 + \left(\sqrt{2} - 1\right) + 2\left(\frac{3}{2} - \sqrt{2}\right)$$

$$= 2 - \sqrt{2}$$

$\Rightarrow \alpha = 2$

$\Rightarrow 6\alpha^3 = 48$

96. If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, where $a, b \in N$, then $a + b$ is equal to _____.

[JEE(Main)April,2024]

यदि $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, जहाँ $a, b \in N$ है, तो $a + b$ बराबर है _____

[JEE(Main)April,2024]

Ans. 8

Sol. $\ell = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{\sin^2 x + \cos^2 x + \sin x \cos x} dx$

$$\ell = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x}{1 + \tan x + \tan^2 x} dx = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x \cdot \sec^2 x dx}{\left(1 + \tan^2 x\right)\left(1 + \tan x + \tan^2 x\right)}$$

Let $\tan x = t$

$$\ell = \int_0^1 \frac{t^2}{\left(1 + t^2\right)\left(1 + t + t^2\right)} dt$$

$$= \int_0^1 \left(\frac{x}{1 + x^2} - \frac{x}{1 + x + x^2}\right) dx$$

$$= \frac{1}{2} \int_0^1 \frac{2x}{1 + x^2} dx - \int_0^1 \frac{\frac{1}{2}(2x + 1) - \frac{1}{2}}{1 + x + x^2} dx$$

$$= \frac{1}{2} \ell n 2 - \frac{1}{2} \ln 3 + \frac{1}{2} \int_0^1 \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{2}\cdot\frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2x+1}{\sqrt{3}}\right]_0^1$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\cdot\frac{\pi}{6}$$

$\therefore$ a = 2, b = 6

$\therefore$ a + b = 8

97. If $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$, then the value of $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ equals _____.

[JEE(Main)April,2024]

यदि $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$ है, तो $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ का मान है _____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $f(t) = \int_0^{\pi} \frac{2xdx}{1-\cos^2 t\sin^2 x}$

$x \to \pi - x$

$$f(t) = \int_0^{\pi} \frac{2(\pi-x)dx}{1-\cos^2 t\sin^2 x} = 2\pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x} - f(t)$$

$$\Rightarrow f(t) = \pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$= 2\pi\int_0^{\frac{\pi}{2}} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$f(t) = 2\pi\int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t\tan^2 x}$$

$$I_1 = \int \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t\tan^2 x}$$

Put cost tanx = $\lambda \Rightarrow$ cost $\sec^2 x dx = d\lambda$

$$I_1 = \int \frac{d\lambda}{\cos t.\left(1+\lambda^2 \sec^2 t - \lambda^2\right)} = \int \frac{d\lambda}{\cos t\left(1+\lambda^2 \tan^2 t\right)}$$

$$= \frac{1}{\cos t.\tan^2 t}.\int \frac{d\lambda}{\lambda^2 + \cos^2 t} = \frac{1}{\cos t \tan^2 t} \times \frac{1}{\cos t} \tan^{-1}\left(\lambda \tan t\right)$$

$$= \frac{1}{\sin t} \tan^{-1}\left(\sin t \tan x\right)$$

$$\Rightarrow f(t) = \frac{2\pi}{\sin t} \tan^{-1}\left(\sin t \tan x\right)\Big]_0^{\frac{\pi}{2}} = \frac{\pi^2}{\sin t}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)} = \int_0^{\frac{\pi}{2}} \sin t dt = 1$$

98. Let $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$. Then the value of $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ is equal to _____.

[JEE(Main)April,2024]

माना $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$ है। तो $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ का मान बराबर है _____

[JEE(Main)April,2024]

Ans. 65

Sol. $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$

$$\int_0^1 \left(1-x^7\right)^{k+1} dx = \int_0^1 \left(1-x^7\right)^k \cdot \left(1-x^7\right) dx$$

$$= \int_0^1 \left(1-x^7\right)^k dx - \int_0^1 x^7 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - \int_0^1 x \cdot x^6 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\int_0^1 x^6 \left(1-x^7\right)^k dx + \int_0^1 \left[1 \cdot \int x^6 \cdot \left(1-x^7\right)^k dx\right] dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\frac{\left(1-x^7\right)^{k+1}}{-7(k+1)}\Bigg|_0^1 + \int_0^1 \frac{\left(1-x^7\right)^{k+1}}{-7(k+1)} dx$$

$$\Rightarrow \int_0^1 \left(1-x^7\right)^{k+1} dx \left\{1+\frac{1}{7(k+1)}\right\} = \int_0^1 \left(1-x^7\right)^k dx - 0$$

$$\frac{7k+8}{7(k+1)} = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$$

$$\sum_{k=1}^{10} \frac{1}{7\left(r_k-1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} \frac{1}{7\left(1+\frac{1}{7k+7}-1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} (K+1) \qquad \Rightarrow \frac{10\times 11}{2}+10 \Rightarrow 55+10=65$$

99. Let [t] denote the largest integer less than or equal to t. If

$\int_0^3 \left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx = a+b\sqrt{2}-\sqrt{3}-\sqrt{5}+c\sqrt{6}-\sqrt{7}$, where a, b, c $\in$ Z, then a + b + c is equal to _____.

[JEE(Main)April,2024]

माना [t], t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। यदि

$\int_0^3 \left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx = a+b\sqrt{2}-\sqrt{3}-\sqrt{5}+c\sqrt{6}-\sqrt{7}$, जहाँ a, b, c $\in$ Z तो, a + b + c बराबर है _____

[JEE(Main)April,2024]

Ans. 23

Sol. $\int_0^3 \left(\left[x^2\right]+\int_0^3 \left[\frac{x^2}{2}\right]\right)dx$

$$= \int_0^1 0dx + \int_1^{12} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx$$

$$+ \int_{\sqrt{3}}^{2} 3dx + \int_2^{\sqrt{5}} 4dx + \int_{\sqrt{5}}^{\sqrt{6}} 5dx + \int_{\sqrt{6}}^{\sqrt{7}} 6dx + \int_{\sqrt{7}}^{\sqrt{8}} 7dx + \int_{\sqrt{8}}^{3} 8dx + \int_0^{\sqrt{2}} 0dx + \int_{\sqrt{2}}^{2} 1dx$$

$$+ \int_2^{\sqrt{6}} 2dx + \int_{\sqrt{6}}^{\sqrt{8}} 3dx + \int_{\sqrt{8}}^{3} 4dx = 31-6\sqrt{2}-\sqrt{3}-\sqrt{5}-2\sqrt{5}\sqrt{6}-\sqrt{7}$$

a = 31 b = –6, c = –2

a + b + c = 31 – 6 – 2 = 23

100. Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8}{\left(n^2+4\right)\sqrt{n^4+16}}+....+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

[JEE(Main)April,2024]

माना $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8}{\left(n^2+4\right)\sqrt{n^4+16}}+....+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right)=\frac{\pi}{k}$ है, जब प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मान लिए गए है। तो k^2 बराबर है _____

[JEE(Main)April,2024]

Ans. 32

Sol. $\sum\limits_{r=1}^{\infty}\frac{n}{\sqrt{n^4+r^4}}-\frac{2nr^2}{\left(n^2+r^2\right)\sqrt{n^4+r^4}}$

$\sum\limits_{r=1}^{\infty}\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}}-\frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$

$\Rightarrow\int\limits_0^1\frac{dx}{\sqrt{1+x^4}}-\frac{2x^2dx}{\left(1+x^2\right)\sqrt{1+x^4}}$

$\Rightarrow\int\limits_0^1\frac{1-x^2}{\left(1+x^2\right)\sqrt{1+x^4}}dx$

$$\Rightarrow \int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\Rightarrow -\int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

$$x+\frac{1}{x}=t \Rightarrow 1-\frac{1}{x^2}dx=dt$$

$$\Rightarrow -\int_\infty^2 \frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow -\int_\infty^2 \frac{tdt}{t^2\sqrt{t^2-2}}$$

take $t^2 - 2 = \alpha^2$

$tdt = ad\alpha$

$$\Rightarrow -\int_\infty^{\sqrt{2}} \frac{\alpha d\alpha}{\left(\alpha^2+2\right)\alpha}$$

$$\Rightarrow -\int_\infty^{\sqrt{2}} \frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\Big]_\infty^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\{\tan^{-1}\}+\frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left\{\frac{\pi}{2}-\frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\pi}{\sqrt{2}}=\frac{\pi}{K}$$

So, $K = 4\sqrt{2}$

$K^2 = 32$